Type MS · Deriving the double-angle formulae

Answer 1 of 24 · Question 11 part a · 3 marks

2021 JUN · Paper 01 · extract 1 of 5 · find part a in this mark scheme extract

Number		
11 (a)	$\cos 2x = \cos(x+x)$	
	$= \cos^2 x - \sin^2 x \ = 1 - \sin^2 x - \sin^2 x$	M1 M1
	$= 1 - 2\sin^2 x$ *	A1 cso (3)
(b)	$\sin x + 2\sin^2 x - 1 = 0 \Rightarrow \ (2\sin x - 1)(\sin x + 1) = 0$	M1dM1
	When $\sin x = \frac{1}{2}$ When $\sin x = -1$	
	$x = \frac{\pi}{6}, \frac{5\pi}{6}$ $x = \frac{3\pi}{2}$	A1
	$\left(\frac{\pi}{6}, \frac{5}{2}\right), \left(\frac{5\pi}{6}, \frac{5}{2}\right), \left(\frac{3\pi}{2}, 1\right)$	A1 (4)
(c)	$\int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (\sin x + 2)\,dx - \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (\cos 2x + 2)\,dx$	M1
	$\left[-\cos x + 2x\right]_{\frac{\pi}{6}}^{\frac{5\pi}{6}} - \left[\frac{\sin 2x}{2} + 2x\right]_{\frac{\pi}{6}}^{\frac{5\pi}{6}}$	A1
	$\left[\left(-\cos\frac{5\pi}{6} + 2\left(\frac{5\pi}{6}\right)\right) - \left(-\cos\frac{\pi}{6} + 2\left(\frac{\pi}{6}\right)\right)\right]$	
	$-\left[\left(\frac{\sin 2\left(\frac{5\pi}{6}\right)}{2} + 2\left(\frac{5\pi}{6}\right)\right) - \left(\frac{\sin 2\left(\frac{\pi}{6}\right)}{2} + 2\left(\frac{\pi}{6}\right)\right)\right]$	M1
	$= \frac{3\sqrt{3}}{2}$	A1
	$\int_{\frac{5\pi}{6}}^{\frac{3\pi}{2}} (\cos 2x + 2)\,dx - \int_{\frac{5\pi}{6}}^{\frac{3\pi}{2}} (\sin x + 2)\,dx$	M1
	$\left[\frac{\sin 2x}{2} + 2x\right]_{\frac{5\pi}{6}}^{\frac{3\pi}{2}} - \left[-\cos x + 2x\right]_{\frac{5\pi}{6}}^{\frac{3\pi}{2}}$	
	$\left[\left(\frac{\sin 2\left(\frac{3\pi}{2}\right)}{2} + 2\left(\frac{3\pi}{2}\right)\right) - \left(\frac{\sin 2\left(\frac{5\pi}{6}\right)}{2} + 2\left(\frac{5\pi}{6}\right)\right)\right]$	M1
	$-\left[\left(-\cos\left(\frac{3\pi}{2}\right) + 2\left(\frac{3\pi}{2}\right)\right) - \left(-\cos\left(\frac{5\pi}{6}\right) + 2\left(\frac{5\pi}{6}\right)\right)\right]$	
	$\frac{3\sqrt{3}}{4}$	A1
	$R_1 : R_2 = 2 : 1$	B1ft
	See Special Case at end of notes	(8)
		Total 15 marks

Type MS · Deriving the double-angle formulae

Answer 1 of 24 · Question 11 part a · 3 marks

2021 JUN · Paper 01 · extract 2 of 5 · find part a in this mark scheme extract

Part	Mark	Notes
(a)	M1	For using the addition formula on page 2 and reaching the result $\cos 2x = \cos^2 x - \sin^2 x$ This **must** be correct for this mark.
	M1	For using **and** applying $\sin^2 x + \cos^2 x = 1$ in their $\cos 2x$ $\cos 2x = 1 - \sin^2 x - \sin^2 x$
	A1 cso	For the correct identity as shown in full. $\cos 2x = 1 - 2\sin^2 x$ * **Note: This is a show question, there must be no errors seen.**
(b)	M1	Equates the two given equations together. $\sin x + 2 = \cos 2x + 2$ and uses the given identity in part (a) to attempt to form a 3TQ in sin x: $\sin x + 2\sin^2 x - 1 = 0$ or $2\sin^2 x + \sin x - 1 = 0$ At least either $2\sin^2 x$ or $\sin x$ must be correct.
	dM1	Solves their 3TQ by any method to reach two values for $\sin x$ -see General Guidance $(2\sin x - 1)(\sin x + 1) = 0 \Rightarrow \sin x = \left(-1,\ \frac{1}{2}\right)$ This mark is dependent on the previous M mark.
	A1	For finding all three values of x: $x = \frac{\pi}{6},\ \frac{5\pi}{6},\ \frac{3\pi}{2}$ Allow $x = 30^\circ,\ 150^\circ,\ 270^\circ$ for this mark – condone missing degree signs
	A1	For all three sets of correct coordinates: $\left(\frac{\pi}{6}, \frac{5}{2}\right), \left(\frac{5\pi}{6}, \frac{5}{2}\right), \left(\frac{3\pi}{2}, 1\right)$ Allow $\left(30^\circ, \frac{5}{2}\right), \left(150^\circ, \frac{5}{2}\right), \left(270^\circ, 1\right)$ – condone missing degree signs
(c)	**Area R_1**	
	M1	For stating a correct method to find the area R_1 using their limits from part (b) of $\frac{\pi}{6}$ and $\frac{5\pi}{6}$ applied the correct way. Condone angles in degrees. $\int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (\sin x + 2)\,dx - \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (\cos 2x + 2)\,dx$ (The + 2 – 2 may have been simplified).
	M1	For an attempt to integrate their expression for R_1 which must involve $\sin x$ and $\cos 2x$ A minimally acceptable integral of $\sin x \to -\cos x$ **or** $\cos 2x \to \pm\frac{\sin 2x}{2}$ and $2 \to 2x$ (The $2 \to 2x$ may not be present if they've simplified first). At least one trig term must be acceptable **with** $2 \to 2x$ (if present) correct. $\left[-\cos x + 2x\right]_{\frac{\pi}{6}}^{\frac{5\pi}{6}} - \left[\frac{\sin 2x}{2} + 2x\right]_{\frac{\pi}{6}}^{\frac{5\pi}{6}}$ Ignore incorrect, angles in degrees or even absent limits for this mark.

Type MS · Deriving the double-angle formulae

Answer 1 of 24 · Question 11 part a · 3 marks

2021 JUN · Paper 01 · extract 3 of 5 · find part a in this mark scheme extract

	M1	Substitutes their limits correctly into their integrated expression. Must involve a changed expression. $\left[\left(-\cos\frac{5\pi}{6}+2\left(\frac{5\pi}{6}\right)\right)-\left(-\cos\frac{\pi}{6}+2\left(\frac{\pi}{6}\right)\right)\right]-$ $\left[\left(\frac{\sin 2\left(\frac{5\pi}{6}\right)}{2}+2\left(\frac{5\pi}{6}\right)\right)-\left(\frac{\sin 2\left(\frac{\pi}{6}\right)}{2}+2\left(\frac{\pi}{6}\right)\right)\right]$ **Do not accept a substitution in degrees – it must be in radians.** **If the work for the first two method marks to find the area of R_1 is fully correct, this mark can be implied by $\frac{3\sqrt{3}}{2}$. If the student does not get $\left[-\cos x+2x\right]_{\frac{\pi}{6}}^{\frac{5\pi}{6}}-\left[\frac{\sin 2x}{2}+2x\right]_{\frac{\pi}{6}}^{\frac{5\pi}{6}}$ from the first two methods marks, they must show the clear substitution of their limits to be awarded this mark.**
	A1	For the correct area.
Area R_2		
	M1	For an attempt to integrate their expression for R_2 which must involve $\int(\cos 2x+2)\,dx-\int(\sin x+2)\,dx$ or $\int\cos 2x-\sin x\,dx$ if they have simplified the +2 – 2. Ignore incorrect, angles in degrees or even absent limits for this mark. A minimally acceptable integral of $\sin x\to-\cos x$ or $\cos 2x\to\pm\frac{\sin 2x}{2}$ and $2\to 2x$ (The $2\to 2x$ may not be present if they've simplified first). At least one trig term must be acceptable **with** $2\to 2x$ (if present) correct. $\left[\frac{\sin 2x}{2}+2x\right]_{\frac{5\pi}{6}}^{\frac{3\pi}{2}}-\left[-\cos x+2x\right]_{\frac{5\pi}{6}}^{\frac{3\pi}{2}}$
	M1	For substituting their limits correctly into their integrated expression. Must involve a changed expression. $\left[\left(\frac{\sin 2\left(\frac{3\pi}{2}\right)}{2}+2\left(\frac{3\pi}{2}\right)\right)-\left(\frac{\sin 2\left(\frac{5\pi}{6}\right)}{2}+2\left(\frac{5\pi}{6}\right)\right)\right]$ $-\left[\left(-\cos\left(\frac{3\pi}{2}\right)+2\left(\frac{3\pi}{2}\right)\right)-\left(-\cos\left(\frac{5\pi}{6}\right)+2\left(\frac{5\pi}{6}\right)\right)\right]$ **Do not accept a substitution in degrees – it must be in radians.** **If the student's work for the first method mark to find the area of R_2 is fully correct, this M mark can be implied by $\frac{3\sqrt{3}}{4}$. If the student does not get**

Type MS · Deriving the double-angle formulae

Answer 1 of 24 · Question 11 part a · 3 marks

2021 JUN · Paper 01 · extract 4 of 5 · find part a in this mark scheme extract

		$\left[\frac{\sin 2x}{2}+2x\right]_{\frac{5\pi}{6}}^{\frac{3\pi}{2}}-\left[-\cos x+2x\right]_{\frac{5\pi}{6}}^{\frac{3\pi}{2}}$ **from the first method mark, they must show the clear substitution of their limits to be awarded this mark.**
	A1	For the correct area.
Ratio	**area of R_1: area of R_2**	
	B1ft	For a ratio given in its simplest form $a : b$ where a, b are positive integers, ft their values for the area of R_1 *and* R_2 Correct ratio is $R_1:R_2 = 2 : 1$
	SC	**Special case: For the use of degrees in part (c).** **<u>If the area of R_1 and R_2 are both correct</u> when limits in degrees have been substituted** (possible, for example, when the candidate has simplified before integrating): Penalise the first substitution in degrees for the area R_1 by awarding M1 M1 M0 A0. For the area R_2 allow M1 M1 A1 if the substitution (in degrees) and the area of R_2 is correct. B1ft for the correct ratio.

Type MS · Deriving the double-angle formulae

Answer 1 of 24 · Question 11 part a · 3 marks

2021 JUN · Paper 01 · extract 5 of 5 · find part a in this mark scheme extract

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Type MS · Deriving the double-angle formulae

Answer 2 of 24 · Question 10 part a · 2 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 4 · find part a in this mark scheme extract

Number		
10 (a)	$\cos 2A = \cos^2 A - \sin^2 A = \cos^2 A - \left(1 - \cos^2 A\right) = 2\cos^2 A - 1*$	M1 A1cso [2]
(b)	$\cos 4A = 2\cos^2 2A - 1$ $= 2\left(2\cos^2 A - 1\right)^2 - 1$ $= 2\left(4\cos^4 A - 4\cos^2 A + 1\right) - 1$ $= 8\cos^4 A - 8\cos^2 A + 1*$	M1 M1 M1 A1cso [4]
(c)	Let $\frac{\theta}{4} + \frac{\pi}{24} = x$ $\cos^2 x\left(\cos^2 x - 1\right) = \cos^4 x - \cos^2 x$ $\Rightarrow \cos^4 x - \cos^2 x = \frac{\cos 4x - 1}{8}$ $4\left(\frac{\theta}{4} + \frac{\pi}{24}\right) = \theta + \frac{\pi}{6}$ $\frac{\cos\left(\theta + \frac{\pi}{6}\right) - 1}{8} = -\frac{1}{16} \Rightarrow \cos\left(\theta + \frac{\pi}{6}\right) = \frac{1}{2}$ $\Rightarrow \theta + \frac{\pi}{6} = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3} \Rightarrow \theta = \frac{\pi}{6}, \frac{3\pi}{2}$	M1 M1 M1 A1 [5]
Alternative (with x defined as above)		
(c)	$\cos^4 x - \cos^2 x = -\frac{1}{16} \Rightarrow 8\cos^4 x - 8\cos^2 x + 1 = 1 - \frac{8}{16}$ $\cos 4x = \frac{1}{2} \Rightarrow 4x = \theta + \frac{\pi}{6} = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \cdots \Rightarrow \theta = \frac{\pi}{6}, \frac{3\pi}{2}$	M1 M1M1 M1A1
Alternative (with x defined as above)		
(c)	$16\left(\cos^2 x\right)^2 - 16\cos^2 x + 1 = 0$ $\cos^2 x = \frac{-(-16) \pm \sqrt{(-16)^2 - 4\times16\times1}}{2\times16} = \frac{2 \pm \sqrt{3}}{4}$ $\cos x = \sqrt{\frac{2 \pm \sqrt{3}}{4}}$ $(= \pm 0.9659...$ or $\pm 0.2588...)$ $x = 0.2617..., 1.3089..., 1.8325..., 2.8787...\left(= \frac{\pi}{12}, \frac{5\pi}{12}, \frac{7\pi}{12}, \frac{11\pi}{12}...\right)$ $\theta = 4\left(x - \frac{\pi}{24}\right) = \frac{\pi}{6}, \frac{3\pi}{2}$	M1 M1 M1 M1A1
(d)	$f(A) = \frac{\cos 4A}{2} + \frac{1}{2}$ $\left[\frac{\sin 4A}{8} + \frac{A}{2}\right]_{\frac{\pi}{6}}^{\frac{\pi}{2}} = \left(\frac{\sin 2\pi}{8} + \frac{\pi}{4}\right) - \left(\frac{\sin\frac{2\pi}{3}}{8} + \frac{\pi}{12}\right) = \frac{\pi}{4} - \frac{\sqrt{3}}{16} - \frac{\pi}{12} = \frac{\pi}{6} - \frac{\sqrt{3}}{16}$	M1 dM1 ddM1 A1 [4]
		Total 15 marks

Type MS · Deriving the double-angle formulae

Answer 2 of 24 · Question 10 part a · 2 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 4 · find part a in this mark scheme extract

(a)	M1	Use $\cos(A+A)=\cos A\cos A-\sin A\sin A$ (may go straight to $\cos 2A=\cos^2 A-\sin^2 A$) and then try to replace $\sin^2 A$ by $1-\cos^2 A$ Allow with any other single variable.
	A1 cso	Obtain $\cos 2A=2\cos^2 A-1$ with no errors in the working. Must be stated with A as the variable.
(b)		Allow with any other single variable for all M marks. Allow one sign error removing brackets and ft for all M marks.
	M1	$\cos 4A=2\cos^2 2A-1$ or $\cos 4A=\cos^2 2A-\sin^2 2A$ Obtain an expression in terms of cos 2A and/or sin 2A.
	M1	eg $\cos 4A=2\left(2\cos^2 A-1\right)^2-1$, $\cos 4A=\left(2\cos^2 A-1\right)^2-4\left(1-\cos^2 A\right)\cos^2 A$ Obtain a correct expression in cos A. Eliminate terms in sin A and sin 2A.
	M1	Expand brackets. Numerical factors may remain. e.g. $\cos 4A=2\left(4\cos^4 A-2\cos^2 A-2\cos^2 A+1\right)-1$ e.g. $\cos 4A=4\cos^4 A-4\cos^2 A+1-4\cos^2 A+4\cos^4 A$
	A1 cso	Obtain the given result with no errors in the working. Must be stated with A as the variable.
(c)	M1	Expand brackets and use the result from part (b) to express the LHS of the equation in terms of $4x$, $\frac{\cos 4x-1}{8}$, where $x=\frac{\theta}{4}+\frac{\pi}{24}$ **Alternative** Arrange the equation in the form $8\cos^4 x-8\cos^2 x+1=1-\frac{8}{16}$ Allow a single numerical or sign error in the constants for both methods.
	M1	Rearrange the equation to give $\cos\left(\theta+\frac{\pi}{6}\right)=\frac{1}{2}$ or oe Note that this mark is shown as B1 on ePEN.
	M1	Find any correct positive value for $4\left(\frac{\theta}{4}+\frac{\pi}{24}\right)$, given in radians. eg $\frac{\pi}{3}$
	M1	Find any correct positive value for θ, given in radians. eg $\frac{\pi}{6}$
	A1	Both correct answers with no extras, given in radians as multiples of π. Ignore answers outside the given range.

Type MS · Deriving the double-angle formulae

Answer 2 of 24 · Question 10 part a · 2 marks

2019 JUN · Paper 02 (regional R) · extract 3 of 4 · find part a in this mark scheme extract

Alternative		
(c)	M1	Write the equation as a three term quadratic. e.g. $16\left(\cos^2 x\right)^2 - 16\cos^2 x + 1 = 0, \left(x = \frac{\theta}{4} + \frac{\pi}{24}\right)$, $A^2 - A + \frac{1}{16} = 0, \left(A = \cos^2\left(\frac{\theta}{4} + \frac{\pi}{24}\right)\right)$
	M1	Any valid method to solve their quadratic equation. Note that this mark is shown as B1 on ePEN.
	M1	Any correct value for $x = \frac{\theta}{4} + \frac{\pi}{24}$ in the range $0 \le x \le \pi$, given in radians, either as a decimal or as a multiple of π.
	M1	Either value of θ given in radians, $\theta = \frac{\pi}{6}$ or $\theta = \frac{3\pi}{2}$ or $\theta = 0.523...$ or $\theta = 4.712...$
	A1	Both $\theta = \frac{\pi}{6}$ and $\theta = \frac{3\pi}{2}$ given as multiples of π.
(d)	M1	Use result from part (b) to write $\mathrm{f}(A) = k(\cos 4A \pm 1)$.
	dM1	Integrate their $\mathrm{f}(A)$ correctly, dep on M1 awarded. Ignore limits.
	ddM1	Substitute both given limits, dep on both M marks awarded.
	A1	$\frac{\pi}{6} - \frac{\sqrt{3}}{16}$ *or* $\frac{1}{6}\pi - \frac{1}{16}\sqrt{3}$ simplified in this format. Do not isw.

Type MS · Deriving the double-angle formulae

Answer 2 of 24 · Question 10 part a · 2 marks

2019 JUN · Paper 02 (regional R) · extract 4 of 4 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Deriving the double-angle formulae

Answer 3 of 24 · Question 8 part c · 2 marks

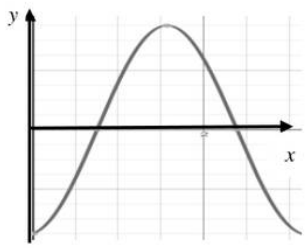
2022 JAN · Paper 02 · extract 1 of 3 · find part c in this mark scheme extract

8(a)	<table><tr><td>x</td><td>0</td><td>0.5</td><td>0.8</td><td>1</td><td>1.6</td><td>2</td><td>2.5</td><td>3</td></tr><tr><td>y</td><td>**–3.5**</td><td>**–1.6**</td><td>0.3</td><td>1.6</td><td>**3.5**</td><td>**2.4**</td><td>–0.8</td><td>–3.2</td></tr></table>	B2
(b)	Their points plotted correctly on the graph Points are joined to form a smooth curve	B1ft B1ft [2]
(c)	$\cos(A+B)=\cos A\cos B-\sin A\sin B \Rightarrow \cos 2A=\cos^2 A-\sin^2 A=1-\sin^2 A-\sin^2 A$ $\cos(2A)=1-2\sin^2 A$	M1 A1 cso [2]
(d)	$2\sin x+6(1-\cos 2x)-x-5=0$ $\sin x-3\cos 2x-\frac{1}{2}=\frac{x}{2}\pm k$ where k is a constant $\Rightarrow y=\frac{x}{2}-1$ Draws line $y=\frac{x}{2}-1$ $x=0.6$ or $x=0.7$ **AND** $x=2.3$ or $x=2.4$	M1 dM1A1 M1 A1 [5]
	Total	**11 marks**

Type MS · Deriving the double-angle formulae

Answer 3 of 24 · Question 8 part c · 2 marks

2022 JAN · Paper 02 · extract 2 of 3 · find part c in this mark scheme extract

<table>
<tr><th>Part</th><th>Mark</th><th>Scheme</th></tr>
<tr><td>(a)</td><td>B2</td><td>
<table>
<tr><td>x</td><td>0</td><td>0.5</td><td>0.8</td><td>1</td><td>1.6</td><td>2</td><td>2.5</td><td>3</td></tr>
<tr><td>y</td><td>–3.5</td><td>–1.6</td><td>0.3</td><td>1.6</td><td>3.5</td><td>2.4</td><td>–0.8</td><td>–3.2</td></tr>
</table>
For all 4 values correct OR If B2 not scored then award B1 for any 2 values correct
Note: These values must be rounded to one decimal place only. Accept only the above values.
</td></tr>
<tr><td>(b)</td><td>B1ft
B1ft</td><td>


All points plotted correctly (ft their values) on the graph within half of one square.
All of their points are joined to form a smooth curve within half of a square of a point. There must be at least six points plotted and joined.
</td></tr>
<tr><td rowspan="2">(c)</td><td>M1</td><td>For using the identity $\cos(A+B)=\cos A\cos B-\sin A\sin B=\cos^2 A-\sin^2 A$ correctly and the identity $\sin^2 A+\cos^2 A=1$ also correctly.</td></tr>
<tr><td>A1
cso</td><td>For simplifying to the required identity $\cos 2A=1-2\sin^2 A$ *
This is a given result so there must be no errors in their solution.</td></tr>
<tr><td rowspan="5">(d)</td><td>M1</td><td>$2\sin^2 x=1-\cos 2x$
For substituting the above into the given f (x) with no errors.
e.g., $2\sin x+6(1-\cos 2x)-x-5=0$</td></tr>
<tr><td>dM1</td><td>For reaching the equation of the curve on one side and the equation of a straight line on the other.
$\sin x-3\cos 2x-\frac{1}{2}=\frac{x}{2}\pm k$ where k is a constant o.e
Note: This mark is dependent on the previous M mark.</td></tr>
<tr><td>A1</td><td>For the correct equation of the equation of the straight line required $y=\frac{x}{2}-1$</td></tr>
<tr><td>M1</td><td>For drawing their line which must be in the form $y=\frac{x}{2}\pm k$
Correct coordinates for $y=\frac{x}{2}-1$ are; $(0,\ -1),\ \left(1,\ -\frac{1}{2}\right),\ (2,\ 0),\ \left(3,\ \frac{1}{2}\right)$</td></tr>
<tr><td>A1</td><td>For both correct roots of $x=0.6$ or $x=0.7$ AND $x=2.3$ or $x=2.4$</td></tr>
</table>

Type MS · Deriving the double-angle formulae

Answer 3 of 24 · Question 8 part c · 2 marks

2022 JAN · Paper 02 · extract 3 of 3 · find part c in this mark scheme extract

Question	Scheme	Mark

Type MS · Deriving the double-angle formulae

Answer 4 of 24 · Question 10 part a.i · 0 marks

2023 JAN · Paper 02 · find part a.i in this mark scheme extract

10 a (i)	$\sin 2\theta = \sin\theta\cos\theta + \cos\theta\sin\theta = 2\sin\theta\cos\theta$ *	M1 A1 cso (2)
a (ii)	$\cos 2\theta = \cos^2\theta - \sin^2\theta = \cos^2\theta - (1-\cos^2\theta)$	M1 M1
	$= 2\cos^2\theta - 1$ *	A1 cso (3)
b	$\sin 2\theta - \tan\theta = 2\sin\theta\cos\theta - \dfrac{\sin\theta}{\cos\theta}$	M1
	$\sin\theta\left(2\cos\theta - \dfrac{1}{\cos\theta}\right)$	M1
	$\sin\theta\left(\dfrac{2\cos^2\theta - 1}{\cos\theta}\right) = \tan\theta\cos 2\theta$ *	dM1 A1cso (4)
c	$\tan x = 0$ so $x = 180$	B1
	$\cos 2x = 0$ so $2x = 90, 270, 450, 630$	M1
	$x = 45, 135, 225, 315$	A1 A1 (4)
		Total 13 marks

Type MS · Deriving the double-angle formulae

Answer 5 of 24 · Question 10 part a.ii · 5 marks

2023 JAN · Paper 02 · find part a.ii in this mark scheme extract

10 a (i)	$\sin 2\theta = \sin\theta\cos\theta + \cos\theta\sin\theta = 2\sin\theta\cos\theta$ *	M1 A1 cso (2)
a (ii)	$\cos 2\theta = \cos^2\theta - \sin^2\theta = \cos^2\theta - (1-\cos^2\theta)$	M1 M1
	$= 2\cos^2\theta - 1$ *	A1 cso (3)
b	$\sin 2\theta - \tan\theta = 2\sin\theta\cos\theta - \dfrac{\sin\theta}{\cos\theta}$	M1
	$\sin\theta\left(2\cos\theta - \dfrac{1}{\cos\theta}\right)$	M1
	$\sin\theta\left(\dfrac{2\cos^2\theta - 1}{\cos\theta}\right) = \tan\theta\cos 2\theta$ *	dM1 A1cso (4)
c	$\tan x = 0$ so $x = 180$	B1
	$\cos 2x = 0$ so $2x = 90, 270, 450, 630$	M1
	$x = 45, 135, 225, 315$	A1 A1 (4)
		Total 13 marks

Type MS · Deriving the double-angle formulae

Answer 6 of 24 · Question 9 part a.i · 0 marks

2023 JUN · Paper 01 · extract 1 of 4 · find part a.i in this mark scheme extract

Question	Scheme	Marks
9(a)	$\cos 2A = \cos^2 A - \sin^2 A = \cos^2 A - (1-\cos^2 A) = 2\cos^2 A - 1$ OR $\cos 2A = \cos^2 A - \sin^2 A = (1-\sin^2 A) - \sin^2 A = 1 - 2\sin^2 A$ (i) $\cos^2 A = \frac{\cos 2A + 1}{2}$, (ii) $\sin^2 A = \frac{1-\cos 2A}{2}$ *	M1M1 A1A1cso [4]
(b)	$(2\sin x - \cos x)(\sin x - 3\cos x)$ $= 2\sin^2 x - 7\sin x\cos x + 3\cos^2 x$ $= 2\left(\frac{1-\cos 2x}{2}\right), \ldots\ldots\ldots\ldots, +3\left(\frac{\cos 2x+1}{2}\right)$ $= \ldots\ldots\ldots\ldots - \frac{7\sin 2x}{2} + \ldots\ldots\ldots\ldots$ $= \frac{1}{2}(\cos 2x - 7\sin 2x + 5)$*	M1 M1, M1 M1 A1 cso [5]
(c)	$y = \frac{1}{2}(\cos 2x - 7\sin 2x + 5) \Rightarrow \frac{dy}{dx} = -\sin 2x - 7\cos 2x = 0$ $\Rightarrow -\sin 2x = 7\cos 2x \Rightarrow \tan 2x = -7$ $2x = -81.869, \quad 98.130, \quad 278.130$ $\Rightarrow x = 49, \quad 139$	M1 M1 A1 A1 [4]
		Total 13 marks

Part	Mark	Notes
NB: Parts (a) and (b) are proofs with given answers. You must check every line of their working.		
(a)	M1	Uses the formula for $\cos(A+B) = \cos A\cos B - \sin A\sin B$ correctly.
	M1	Uses the identity $\cos^2 A = 1 - \sin^2 A$ or $\sin^2 A = 1 - \cos^2 A$
(i)	A1 cso	For the correct identity for either $\sin^2 A$ or $\cos^2 A$ with no errors seen
(ii)	A1 cso	For the correct identity for $\sin^2 A$ and $\cos^2 A$ with no errors seen
SC	If a candidate does not use $\cos(A+B) = \cos A\cos B - \sin A\sin B$ to find the required identities for $\sin^2 A$ or $\cos^2 A$ but uses them to show $\sin^2 A = \frac{1-\cos 2A}{2}$ or $\cos^2 A = \frac{\cos 2A+1}{2}$ apply the following scheme.	
	First M mark – M0	
	2nd M1	Uses the identity $\cos^2 A = 1 - \sin^2 A$ or $\sin^2 A = 1 - \cos^2 A$ For example:

Type MS · Deriving the double-angle formulae

Answer 6 of 24 · Question 9 part a.i · 0 marks

2023 JUN · Paper 01 · extract 2 of 4 · find part a.i in this mark scheme extract

		$\cos^2 A = \frac{\cos 2A + 1}{2} = 1 - \sin^2 A$ $\sin^2 A = 1 - \frac{\cos 2A + 1}{2} = \frac{2 - (\cos 2A + 1)}{2} = \frac{1 - \cos 2A}{2}$
(i)/(ii)	**A1 cso**	For the correct identity for $\sin^2 A$ or $\cos^2 A$ with no errors seen
	A0	Not available
(b)	**M1**	Multiplies out the brackets with no errors.
	M1	Substitutes at least one of the given identities for $\cos 2x$ into their expansion [either for $\sin^2 x$ or $\cos^2 x$]. This is not a dependent mark, allow for substitution into an erroneous expression.
	M1	Substitutes both of the given identities for $\cos 2x$ into their expansion [for both $\sin^2 x$ and $\cos^2 x$]. This is not a dependent mark, allow for substitution into an erroneous expression. **NB: This is a B mark in Epen**
	M1	Uses the identity $\sin 2A = 2\sin A \cos A$ seen applied explicitly **before** the final answer. Look for, $k \sin x \cos x \rightarrow \frac{k \sin 2x}{2}$
	A1 cso	For the correct identity with no errors seen.
	ALT – Works backwards – these steps must contain no errors (to maintain compatibility with the main method. Starting from $\frac{1}{2}(\cos 2x - 7\sin 2x + 5)$ award as follows:	
	M1	Uses: $\cos 2x = \cos^2 x - \sin^2 x$
	M1	Uses: $5 = 5\cos^2 x + 5\sin^2 x$
	M1	Uses: $-7\sin 2x = 14 \sin x \cos x$
	M1	Collects like terms and divides by 2 to give: $3\cos^2 x - 7\sin x \cos x + 2\sin^2 x$
	A1	Factorises their expression to give: $(2\sin x - \cos x)(\sin x - 3\cos x)$
(c)	**M1**	For an attempt to differentiate the given equation. A minimally acceptable is defined as follows: $\frac{dy}{dx} = \pm k \sin 2x \pm l \cos 2x$ where k and l are integers **with no other terms**.
	M1	Uses the identity $\tan 2x = \frac{\sin 2x}{\cos 2x}$ This is an independent mark, but it must be used correctly following correct division of $\cos 2x$ throughout.
	A1	Find at least one **correct** value for $2x$ e.g., –81.869, 98.130, 278.130 etc **NB: This is an M mark in Epen.** [Accept – 81, 98, 278]
	A1	For $x = 49$ and 139 rounded to the nearest whole number or better.

Type MS · Deriving the double-angle formulae

Answer 6 of 24 · Question 9 part a.i · 0 marks

2023 JUN · Paper 01 · extract 3 of 4 · find part a.i in this mark scheme extract

	ALT - Differentiates using product rule	
	M1	$y=(2\sin x-\cos x)(\sin x-3\cos x)$ $\frac{dy}{dx}=(2\sin x-\cos x)(\cos x+3\sin x)+(\sin x-3\cos x)(2\cos x+\sin x)$ $\frac{dy}{dx}=2\sin x\cos x+6\sin^2 x-\cos^2 x-3\sin x\cos x+$ $+2\sin x\cos x+\sin^2 x-6\cos^2 x-3\sin x\cos x$ $\frac{dy}{dx}=-2\sin x\cos x+6\sin^2 x-7\cos^2 x$ **Allow** $\frac{dy}{dx}=\pm p\sin x\cos x\pm q\sin^2 x\pm r\cos^2 x$ **NB: This will not factorise**
	M1	Divides through by $\cos^2 x$ provided they have an expression of the form $\frac{dy}{dx}=\pm p\sin x\cos x\pm q\sin^2 x\pm r\cos^2 x$ This gives : $\frac{dy}{dx}=-2\sin x\cos x+6\sin^2 x-7\cos^2 x=0 \quad \rightarrow \div\cos^2 x$ $-7-2\tan x+7\tan^2 x=0$ **AND** solves to give a value for $\tan x=\frac{1\mp 5\sqrt{2}}{7}$ or allow $\tan x=\frac{1\pm 5\sqrt{2}}{7}$
	A1	At least one correct value for x of for example, awrt 49.1….awrt – 40.9
	A1	For both 49° and 139°

Type MS · Deriving the double-angle formulae

Answer 6 of 24 · Question 9 part a.i · 0 marks

2023 JUN · Paper 01 · extract 4 of 4 · find part a.i in this mark scheme extract

Question	Scheme	Marks

Type MS · Deriving the double-angle formulae

Answer 7 of 24 · Question 9 part a.ii · 4 marks

2023 JUN · Paper 01 · find part a.ii in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Deriving the double-angle formulae

Answer 8 of 24 · Question 11 part a · 2 marks

2024 JUN · Paper 02 (regional R) · extract 1 of 3 · find part a in this mark scheme extract

11(a)	$cos(A+A) = cos\,A\,cos\,A - sin\,A\,sin\,A$ $\Rightarrow cos\,2A = cos^2 A - (1 - cos^2 A)$ $cos\,2A = 2\,cos^2 A - 1$ *	M1 A1 cso [2]
(b)	$(2\cos^2 A - 1)^2 = (\cos 2A)^2$ $\cos 4A = 2\cos^2 2A - 1 \Rightarrow \cos^2 2A = \frac{\cos 4A + 1}{2}$ $(2\cos^2 2A - 1)^2 = \frac{\cos 4A + 1}{2}$ *	M1 M1 A1 cso [3]
(b) **ALT**	$cos\,4A = 2\,cos^2 2A - 1$ $= 2(2\,cos^2 A - 1)^2 - 1$ $(2\,cos^2 A - 1)^2 = \frac{cos\,4A + 1}{2}$ *	M1 M1 A1 cso [3]
(c)	$y = \frac{sin\,2x}{2} + \frac{(2\,cos^2 x - 1)^2}{2} + \frac{1}{8} \Rightarrow y = \frac{sin\,2x}{2} + \frac{cos\,4x}{4} + \frac{3}{8}$ $\frac{dy}{dx} = cos\,2x - sin\,4x$ $\frac{dy}{dx} = cos\,2x - 2\,sin\,2x\,cos\,2x \{= 0\}$ $\{cos\,2x(1 - 2\,sin\,2x) = 0\} \Rightarrow sin\,2x = \frac{1}{2}$ $\Rightarrow x = \frac{\sin^{-1}\left(\frac{1}{2}\right)}{2} = \cdots$ $x = \frac{\pi}{12}$ $y = \frac{sin2\left("\frac{\pi}{12}"\right)}{2} + \frac{cos4\left("\frac{\pi}{12}"\right)}{4} + \frac{3}{8} = \frac{3}{4} \{\Rightarrow \left(\frac{\pi}{12}, \frac{3}{4}\right)\}$	M1 dM1 B1 ddM A1 M1A1 [7]
		Total 12 marks

Type MS · Deriving the double-angle formulae

Answer 8 of 24 · Question 11 part a · 2 marks

2024 JUN · Paper 02 (regional R) · extract 2 of 3 · find part a in this mark scheme extract

Part	Mark	Notes
(a)	M1	For correct use of $cos(A+B) = cos\,A\,cos\,B - sin\,A\,sin\,B \Rightarrow$ $cos(A+A) = cos\,A\,cos\,A - sin\,A\,sin\,A$ ***OR*** $cos\,2A = cos^2 A - sin^2 A$ **with** the Pythagorean identity to eliminate sine from the identity.
	A1 cso	Achieves printed answer with no errors seen. Left hand side must be cos2A not cos(A+A)
(b)	M1	Uses the given result from (a), replaces $(2\cos^2 A - 1)^2$ with $(\cos 2A)^2$
	M1	Uses the given result from (a) or a correct identity to replace $\cos^2 2A$ with $\frac{\cos 4A+1}{2}$
	A1 cso	Fully correct proof showing all necessary steps. (If they meet in the middle, they must have a conclusion, accept e.g. shown, proved, L=R, etc.)
(b) ALT	M1	Uses the given result from (a) on cos 4A
	M1	Uses the given result from (a) or a correct identity on cos 2A
	A1 cso	Fully correct proof showing all necessary steps. (If they meet in the middle, they must have a conclusion, accept e.g. shown, proved, L=R, etc.)
(c)	M1	Rearranges the given y into the form $\frac{sin\,2x}{2} + \frac{cos\,4x+1}{k} + l$, oe, where k and l are nonzero constants Or $\frac{sin\,2x}{2} + \frac{cos\,4x}{P} + Q$, oe, where P and Q are nonzero constants (Allow mixed variables x and A for this method mark only)
	dM1	For an attempt to differentiate their y into the form $P\cos(2x) \pm Q\sin(4x)$ *oe, where P, Q are nonzero constants* **Depends on the previous method mark.**
	B1	Uses $\sin 4x = 2\sin 2x\cos 2x$ or $\sin 4x = \sqrt{1-\cos^2 4x}$ oe in their dy/dx (may be implied by correct working)
	ddM1	Sets their differentiated expression = 0 (can be implied) and attempts a correct method to solve for x=... **Depends on both previous method marks.**
	A1	For $x = \frac{\pi}{12}$, ignore extra x values are outside of given range. Withhold A mark if extra solution seen in the given range.
	M1	For attempting to find the value of y using their x, found by solving their $\frac{dy}{dx} = 0$, provided x is in the range $0 \leqslant x \leqslant \frac{\pi}{6}$ Substitutes into the given y **or** their y, correct answer implies this mark, but for incorrect answer, must see explicit substitution
	A1	For the correct exact coordinates of P, $\left(\frac{\pi}{12}, \frac{3}{4}\right)$ *or* $\left(\frac{\pi}{12}, 0.75\right)$, accept $x = \frac{\pi}{12}$, $y = \frac{3}{4}$ *or* $x = \frac{\pi}{12}$, $y = 0.75$

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Answer 8 of 24 · Question 11 part a · 2 marks

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Answer 9 of 24 · Question 11 part a.i · 3 marks

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11(a)(i)	$\cos 2A = \cos^2 A - \sin^2 A, = \cos^2 A - \left(1 - \cos^2 A\right)$ $\cos 2A = 2\cos^2 A - 1$ *	M1,M1 A1 cso [3]
(ii)	$\sin 2A = \sin A\cos A + \sin A\cos A = 2\sin A\cos A$ *	B1 cso [1]
(b)	$\cos 3A = \cos\left(2A + A\right)$ $= \cos 2A\cos A - \sin A\sin 2A$ $= \left(2\cos^2 A - 1\right)\cos A - 2\sin A\sin A\cos A$ $= 2\cos^3 A - \cos A - 2\left(1 - \cos^2 A\right)\cos A$ $= 2\cos^3 A - \cos A + 2\cos^3 A - 2\cos A$ $= 4\cos^3 A - 3\cos A$ $\Rightarrow \cos^3 A = \dfrac{\cos 3A + 3\cos A}{4}$ * **ALT** $\cos^3 A = \dfrac{\cos 3A + 3\cos A}{4}$ $= \dfrac{\cos 2A\cos A - \sin 2A\sin A + 3\cos A}{4}$ $= \dfrac{\left(2\cos^2 A - 1\right)\cos A - 2\sin A\sin A\cos A + 3\cos A}{4}$ $= \dfrac{2\cos^3 A - \cos A - 2\left(1 - \cos^2 A\right)\cos A + 3\cos A}{4}$ $= \dfrac{2\cos^3 A - \cos A + 2\cos^3 A - 2\cos A + 3\cos A}{4}$ $= \dfrac{4\cos^3 A}{4} = \cos^3 A \quad [LHS = RHS]$	M1 M1 M1 A1 Cso [4] [M1 M1 M1 A1]
(c)	$\left[\cos 3A = 4\cos^3 A - 3\cos A \Rightarrow 2\cos 3A = 8\cos^3 A - 6\cos A\right]$ $8\cos^3\left(\dfrac{\theta}{2}\right) - 6\cos\left(\dfrac{\theta}{2}\right) - 1 = 0 \Rightarrow 2\cos 3\left(\dfrac{\theta}{2}\right) - 1 = 0$ $\cos 3\left(\dfrac{\theta}{2}\right) = \dfrac{1}{2} \Rightarrow 3\left(\dfrac{\theta}{2}\right) = \dfrac{\pi}{3}, \dfrac{5\pi}{3}, \dfrac{7\pi}{3}, \left(\dfrac{11\pi}{3}\right)$ $\theta = \dfrac{2\pi}{9}, \dfrac{10\pi}{9}, \dfrac{14\pi}{9}$	M1 M1 M1A1 [4]

Type MS · Deriving the double-angle formulae

Answer 9 of 24 · Question 11 part a.i · 3 marks

2024 JUN · Paper 02 · extract 2 of 4 · find part a.i in this mark scheme extract

(d)	$I=\int_0^{\frac{\pi}{6}}\left(4\cos^3\theta-\sin 2\theta\right)\mathrm{d}\theta=\int_0^{\frac{\pi}{6}}\left(\cos 3\theta+3\cos\theta-\sin 2\theta\right)\mathrm{d}\theta$	M1
	$I=\left[\frac{\sin 3\theta}{3}+3\sin\theta+\frac{\cos 2\theta}{2}\right]_0^{\frac{\pi}{6}}$	M1
	$=\left[\frac{\sin 3\left(\frac{\pi}{6}\right)}{3}+3\sin\left(\frac{\pi}{6}\right)+\frac{\cos 2\left(\frac{\pi}{6}\right)}{2}\right]-\left[\frac{\sin 3(0)}{3}+3\sin(0)+\frac{\cos 2(0)}{2}\right]$	M1
	$=\left[\frac{1}{3}+\frac{3}{2}+\frac{1}{4}\right]-\left[0-0+\frac{1}{2}\right]=\frac{19}{12}$	A1 [4]
		Total 16 marks

Type MS · Deriving the double-angle formulae

Answer 9 of 24 · Question 11 part a.i · 3 marks

2024 JUN · Paper 02 · extract 3 of 4 · find part a.i in this mark scheme extract

(c)	M1	Substitutes $2\cos 3\left(\frac{\theta}{2}\right)$ in place of $8\cos^3\left(\frac{\theta}{2}\right)-6\cos\left(\frac{\theta}{2}\right)$ correctly to obtain at least $k\cos 3\left(\frac{\theta}{2}\right)-1=0$ where k is a constant
	M1	Rearranges to obtain $\cos 3\left(\frac{\theta}{2}\right)=\frac{1}{k}$ and takes the inverse cosine to find at least any one **correct** value for $3\left(\frac{\theta}{2}\right)$
	M1	Finds at least one correct value for θ
	A1	Finds all three values with no extra values within range. Ignore any values given outside of the range.
(d)	M1	For a **correct** substitution as shown. $4\cos^3\theta=\cos 3\theta+3\cos\theta$
	M1	For an acceptable attempt to integrate either $\cos 3\theta\rightarrow\pm\frac{\sin 3\theta}{3}$ or $-\sin 2\theta\rightarrow\pm\frac{\cos 2\theta}{2}$
	M1	For substituting both given values the correct way around. This mark can be awarded for a complete substitution SEEN into any changed expression. Award this mark for **correct** integration **and** a value of $\frac{19}{12}$ seen without explicit substitution.
	A1	For $\frac{19}{12}$

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Answer 9 of 24 · Question 11 part a.i · 3 marks

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Answer 10 of 24 · Question 11 part a.ii · 1 mark

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11(a)(i)	$\cos 2A = \cos^2 A - \sin^2 A, = \cos^2 A - \left(1 - \cos^2 A\right)$ $\cos 2A = 2\cos^2 A - 1$ *	M1,M1 A1 cso [3]
(ii)	$\sin 2A = \sin A\cos A + \sin A\cos A = 2\sin A\cos A$ *	B1 cso [1]
(b)	$\cos 3A = \cos\left(2A + A\right)$ $= \cos 2A\cos A - \sin A\sin 2A$ $= \left(2\cos^2 A - 1\right)\cos A - 2\sin A\sin A\cos A$ $= 2\cos^3 A - \cos A - 2\left(1 - \cos^2 A\right)\cos A$ $= 2\cos^3 A - \cos A + 2\cos^3 A - 2\cos A$ $= 4\cos^3 A - 3\cos A$ $\Rightarrow \cos^3 A = \dfrac{\cos 3A + 3\cos A}{4}$ * **ALT** $\cos^3 A = \dfrac{\cos 3A + 3\cos A}{4}$ $= \dfrac{\cos 2A\cos A - \sin 2A\sin A + 3\cos A}{4}$ $= \dfrac{\left(2\cos^2 A - 1\right)\cos A - 2\sin A\sin A\cos A + 3\cos A}{4}$ $= \dfrac{2\cos^3 A - \cos A - 2\left(1 - \cos^2 A\right)\cos A + 3\cos A}{4}$ $= \dfrac{2\cos^3 A - \cos A + 2\cos^3 A - 2\cos A + 3\cos A}{4}$ $= \dfrac{4\cos^3 A}{4} = \cos^3 A \quad [LHS = RHS]$	M1 M1 M1 A1 Cso [4] [M1 M1 M1 A1]
(c)	$\left[\cos 3A = 4\cos^3 A - 3\cos A \Rightarrow 2\cos 3A = 8\cos^3 A - 6\cos A\right]$ $8\cos^3\left(\dfrac{\theta}{2}\right) - 6\cos\left(\dfrac{\theta}{2}\right) - 1 = 0 \Rightarrow 2\cos 3\left(\dfrac{\theta}{2}\right) - 1 = 0$ $\cos 3\left(\dfrac{\theta}{2}\right) = \dfrac{1}{2} \Rightarrow 3\left(\dfrac{\theta}{2}\right) = \dfrac{\pi}{3}, \dfrac{5\pi}{3}, \dfrac{7\pi}{3}, \left(\dfrac{11\pi}{3}\right)$ $\theta = \dfrac{2\pi}{9}, \dfrac{10\pi}{9}, \dfrac{14\pi}{9}$	M1 M1 M1A1 [4]

Type MS · Deriving the double-angle formulae

Answer 10 of 24 · Question 11 part a.ii · 1 mark

2024 JUN · Paper 02 · extract 2 of 4 · find part a.ii in this mark scheme extract

(d)	$I = \int_0^{\frac{\pi}{6}} \left(4\cos^3\theta - \sin 2\theta\right) \mathrm{d}\theta = \int_0^{\frac{\pi}{6}} \left(\cos 3\theta + 3\cos\theta - \sin 2\theta\right) \mathrm{d}\theta$	M1
	$I = \left[\frac{\sin 3\theta}{3} + 3\sin\theta + \frac{\cos 2\theta}{2}\right]_0^{\frac{\pi}{6}}$	M1
	$= \left[\frac{\sin 3\left(\frac{\pi}{6}\right)}{3} + 3\sin\left(\frac{\pi}{6}\right) + \frac{\cos 2\left(\frac{\pi}{6}\right)}{2}\right] - \left[\frac{\sin 3(0)}{3} + 3\sin(0) + \frac{\cos 2(0)}{2}\right]$	M1
	$= \left[\frac{1}{3} + \frac{3}{2} + \frac{1}{4}\right] - \left[0 - 0 + \frac{1}{2}\right] = \frac{19}{12}$	A1 [4]
		Total 16 marks

Type MS · Deriving the double-angle formulae

Answer 10 of 24 · Question 11 part a.ii · 1 mark

2024 JUN · Paper 02 · extract 3 of 4 · find part a.ii in this mark scheme extract

(c)	M1	Substitutes $2\cos 3\left(\frac{\theta}{2}\right)$ in place of $8\cos^3\left(\frac{\theta}{2}\right)-6\cos\left(\frac{\theta}{2}\right)$ correctly to obtain at least $k\cos 3\left(\frac{\theta}{2}\right)-1=0$ where k is a constant
	M1	Rearranges to obtain $\cos 3\left(\frac{\theta}{2}\right)=\frac{1}{k}$ and takes the inverse cosine to find at least any one **correct** value for $3\left(\frac{\theta}{2}\right)$
	M1	Finds at least one correct value for θ
	A1	Finds all three values with no extra values within range. Ignore any values given outside of the range.
(d)	M1	For a **correct** substitution as shown. $4\cos^3\theta=\cos 3\theta+3\cos\theta$
	M1	For an acceptable attempt to integrate either $\cos 3\theta\rightarrow\pm\frac{\sin 3\theta}{3}$ or $-\sin 2\theta\rightarrow\pm\frac{\cos 2\theta}{2}$
	M1	For substituting both given values the correct way around. This mark can be awarded for a complete substitution SEEN into any changed expression. Award this mark for **correct** integration **and** a value of $\frac{19}{12}$ seen without explicit substitution.
	A1	For $\frac{19}{12}$

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Answer 10 of 24 · Question 11 part a.ii · 1 mark

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Answer 11 of 24 · Question 8 part a · 1 mark

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8 (a)	$k = 2$	B1 [1]
(b)	$2+3\cos A-\sin A-3\sin 2A-2\cos^2 A=3\cos A-\sin A-6\sin A\cos A+2\sin^2 A$ $3\cos A-\sin A-6\sin A\cos A+2\sin^2 A=3\cos A(1-2\sin A)-\sin A(1-2\sin A)$ $3\cos A(1-2\sin A)-\sin A(1-2\sin A)=(1-2\sin A)(3\cos A-\sin A)$ $\Rightarrow p=3,\ q=1,\ r=2$	B1 M1 A1 [3]
	ALT	
	$2+3\cos A-\sin A-3\sin 2A-2\cos^2 A=3\cos A-\sin A-6\sin A\cos A+2\sin^2 A$ $(p\cos A-\sin A)(q-r\sin A)=pq\cos A-q\sin A-pr\cos A\sin A+r\sin^2 A$ $pq\cos A-q\sin A-pr\cos A\sin A+r\sin^2 A$ $\Rightarrow p=3,\ q=1,\ r=2$	{B1} {M1} {A1} [3]
(c)	$(1-2\sin 2\theta)(3\cos 2\theta-\sin 2\theta)=0$ $\Rightarrow 3\cos 2\theta-\sin 2\theta=0\Rightarrow \tan 2\theta=3$ $\Rightarrow 1-2\sin 2\theta=0\Rightarrow \sin 2\theta=\frac{1}{2}$ $\tan 2\theta=3\Rightarrow 2\theta=1.2490...,\ 4.3906...\ \Rightarrow \theta=0.625,\ 2.20$ $\sin 2\theta=\frac{1}{2}\Rightarrow 2\theta=0.5235...,\ 2.6179...\ \Rightarrow \theta=0.262,\ 1.31\ \left[\frac{\pi}{12},\frac{5\pi}{12}\right]$	B1 B1 M1A1 M1A1 [6]
		Total 10 marks

Type MS · Deriving the double-angle formulae

Answer 11 of 24 · Question 8 part a · 1 mark

2020 JAN · Paper 02 (regional R) · extract 2 of 3 · find part a in this mark scheme extract

(c)	
B1	$\tan 2\theta = 3$
B1	$\sin 2\theta = \frac{1}{2}$
M1	$2\theta = 1.24(90...,)\ 4.39(06...)$
A1	$\theta = 0.625,\ 2.2(0)$
M1	$2\theta = 0.5235...,\ 2.61(79...)$ allow $\left[\frac{\pi}{6}, \frac{5\pi}{6}\right]$
A1	$\theta = 0.262,\ 1.31$ allow $\left[\frac{\pi}{12}, \frac{5\pi}{12}\right]$

Type MS · Deriving the double-angle formulae

Answer 11 of 24 · Question 8 part a · 1 mark

2020 JAN · Paper 02 (regional R) · extract 3 of 3 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Deriving the double-angle formulae

Answer 12 of 24 · Question 1 part a · 2 marks

2021 NOV · Paper 02 · extract 1 of 3 · find part a in this mark scheme extract

1. Factorisation:

$(x^2+bx+c)=(x+p)(x+q)$, where $|pq|=|c|$ leading to $x=....$

$(ax^2+bx+c)=(mx+p)(nx+q)$ where $|pq|=|c|$ and $|mn|=|a|$ leading to $x=....$

2. Formula:

Attempt to use the **correct** formula (shown explicitly or implied by working) with values for *a*, *b* and *c*, leading to $x=....$

3. Completing the square:

$x^2+bx+c=0$: $\left(x\pm\frac{b}{2}\right)^2\pm q\pm c=0, \quad q\neq 0$ leading to $x=....$

4. Use of calculators

Unless the question specifically states 'show' or 'prove' accept correct answers from no working. If an incorrect solution is given without any working do not award the Method mark.

Method marks for differentiation and integration:

1. Differentiation

Power of at least one term decreased by 1. $\left(x^n\rightarrow x^{n-1}\right)$

2. Integration:

Power of at least one term increased by 1. $\left(x^n\rightarrow x^{n+1}\right)$

Type MS · Deriving the double-angle formulae

Answer 12 of 24 · Question 1 part a · 2 marks

2021 NOV · Paper 02 · extract 2 of 3 · find part a in this mark scheme extract

Use of a formula:

Generally, the method mark is gained by **either**

quoting a correct formula and attempting to use it, even if there are mistakes in the substitution of values

or, where the formula is not quoted, the method mark can be gained by implication from the substitution of correct values and then proceeding to a solution.

Answers without working:

The rubric states "Without sufficient working, correct answers may be awarded no marks".

General policy is that if it could be done "in your head" detailed working would not be required. (Mark schemes may override this: eg in a case of "prove or show....")

Exact answers:

When a question demands an exact answer, all the working must also be exact. Once a candidate loses exactness by resorting to decimals the exactness cannot be regained.

Rounding answers (where accuracy is specified in the question)

Penalise only once per question for failing to round as instructed - ie giving more digits in the answers. Answers with fewer digits are automatically incorrect, but the isw rule may allow the mark to be awarded before the final answer is given.

Multiple attempts at a question.

If there is a choice of methods shown, then award the lowest mark, unless the subsequent working makes clear the method that has been used.

Type MS · Deriving the double-angle formulae

Answer 12 of 24 · Question 1 part a · 2 marks

2021 NOV · Paper 02 · extract 3 of 3 · find part a in this mark scheme extract

Question number	Scheme	Marks
1 (a)	$\lfloor \cos 2A = \cos^2 A - \sin^2 A \quad \text{and} \quad \cos^2 A + \sin^2 A = 1 \rfloor$ $\Rightarrow [\cos 2A] = (1 - \sin^2 A) - \sin^2 A$ $\Rightarrow [\cos 2A] = 1 - 2\sin^2 A *$	M1 A1 cso [2]
(b)	$\frac{4-y}{3} = 1 - 2\left(\frac{x+1}{2}\right)^2$ $\Rightarrow \frac{4-y}{3} = 1 - 2\left(\frac{x^2+2x+1}{4}\right) \Rightarrow y = ...$ $\Rightarrow y = \frac{1}{2}(3x^2 + 6x + 5)$ *	M1 M1 A1* cso [3]
		Total 5 marks

Type MS · Deriving the double-angle formulae

Answer 13 of 24 · Question 6 part a · 2 marks

2022 JUN · Paper 02 · extract 1 of 3 · find part a in this mark scheme extract

6 a	$\cos 2A = \cos^2 A - \sin^2 A$ $= (1-\sin^2 A) - \sin^2 A = 1 - 2\sin^2 A$ *	M1 A1 cso (2)
b	$\pi\int_0^{\frac{\pi}{4}} (3+2\sin x)^2 \mathrm{d}x$	M1
	$\pi\int_0^{\frac{\pi}{4}} (9+12\sin x+4\sin^2 x)\,\mathrm{d}x$	M1
	$\pi\int_0^{\frac{\pi}{4}} (11+12\sin x-2\cos 2x)\,\mathrm{d}x$	M1
	$\pi\left[11x-12\cos x-\sin 2x\right]_0^{\frac{\pi}{4}}$	M1
	$\pi\left[\left(\frac{11\pi}{4}-6\sqrt{2}-1\right)-(0-12-0)\right]$ = awrt 35	M1 A1 (6)
		Total 8 marks

Type MS · Deriving the double-angle formulae

Answer 13 of 24 · Question 6 part a · 2 marks

2022 JUN · Paper 02 · extract 2 of 3 · find part a in this mark scheme extract

Part	Mark	Notes
(a)	**M1**	For the use of both $\cos 2A = \cos^2 A - \sin^2 A$ and $\cos^2 A = \left(1 - \sin^2 A\right)$ to attempt to form the required identity.
	A1	For $\cos 2A = 1 - 2\sin^2 A$ with no errors seen.
(b)	**M1**	For the correct use of use of $\pi\int_a^b y^2 \mathrm{d}x$ to obtain: $\pi\int_0^{\frac{\pi}{4}} (3 + 2\sin x)^2 \mathrm{d}x$ with the correct limits and π You may not see π and the limits at this stage. They may add these in at the end of their work. Check and if that is the case, award this mark.
	M1	For expanding the given quadratic correctly to achieve: This must be correct. $\pi\int_0^{\frac{\pi}{4}} \left(9 + 12\sin x + 4\sin^2 x\right) \mathrm{d}x$ or accept $\pi\int_0^{\frac{\pi}{4}} \left(9 + 6\sin x + 6\sin x + 4\sin^2 x\right) \mathrm{d}x$ Ignore limits and π for this mark. **This is an A mark in Epen**
	M1	For using $\sin^2 A = \frac{1 - \cos 2A}{2}$ and substituting this identity into their expansion to attempt to obtain: $\pi\int_0^{\frac{\pi}{4}} (11 + 12\sin x - 2\cos 2x) \mathrm{d}x$ As a minimum we need to see: $A + B\sin x \pm C\cos 2x$ Ignore limits and π for this mark.
	M1	For an attempt to integrate their expression which must be of the form: $A + B\sin x \pm C\cos 2x$ **or** $A + B\sin x \pm C\sin^2 x$ where there must be an acceptable attempt to integrate as a minimum at least **both** A **and** $B\sin x$ to obtain $A \to Ax$ and $B\sin x \to -B\cos x$ Ignore limits and π for this mark.
	M1	For the correct substitution of the correct given limits the correct way around into an integrated/changed expression. $\pi\left[\left(\frac{11\pi}{4} - 6\sqrt{2} - 1\right) - (0 - 12 - 0)\right] = 35.041....$ This mark can be implied by the correct final answer **ONLY** if the correct integration is seen explicitly in the response. If their integrated expression is incorrect, and no substitution is seen award M0 here.
	A1	For awrt 35

Type MS · Deriving the double-angle formulae

Answer 13 of 24 · Question 6 part a · 2 marks

2022 JUN · Paper 02 · extract 3 of 3 · find part a in this mark scheme extract

Question number	Scheme	Marks
	$\tan\theta + \tan\theta$	

Type MS · Deriving the double-angle formulae

Answer 14 of 24 · Question 7 part a · 2 marks

2022 JUN · Paper 02 · extract 1 of 3 · find part a in this mark scheme extract

7(i) (a)	$\tan 2\theta = \dfrac{\tan\theta + \tan\theta}{1 - \tan\theta\tan\theta}$	M1
	$= \dfrac{2\tan\theta}{1-\tan^2\theta}$ *	A1 cso (2)
(b)	$1 = \dfrac{2\tan\alpha}{1-\tan^2\alpha}$	
	$\tan^2\alpha + 2\tan\alpha - 1 = 0$	M1
	$\tan\alpha = -1 \pm \sqrt{2}$	M1 A1 (3)
(ii) (a)	$\cos(x-30) = \sin(x+30)$	
	$\Rightarrow \cos x\cos 30^\circ + \sin x\sin 30^\circ = \sin x\cos 30^\circ + \cos x\sin 30^\circ$	M1
	$\left(\dfrac{\sqrt{3}}{2} - \dfrac{1}{2}\right)\cos x = \left(\dfrac{\sqrt{3}}{2} - \dfrac{1}{2}\right)\sin x$	M1
	$\dfrac{\sin x}{\cos x} = \tan x = \dfrac{\left(\frac{\sqrt{3}}{2} - \frac{1}{2}\right)}{\left(\frac{\sqrt{3}}{2} - \frac{1}{2}\right)}$	M1
	$\tan x = 1$ *	A1 cso (4)
(b)	$\tan 2y = 1 \Rightarrow \tan y = -1 \pm \sqrt{2}$	
	$y = -67.5^\circ, 22.5^\circ$	M1 A1 (2)
		Total 11 marks

Part	Mark	Notes
(i)(a)	**M1**	For use of $\tan(A+B) \Rightarrow \tan 2\theta = \dfrac{\tan\theta+\tan\theta}{1-\tan\theta\tan\theta}$ and simplifying to obtain the given result.
	A1	For the given result with no errors seen
(i)(b)	**M1**	For setting $1 = \dfrac{2\tan\alpha}{1-\tan^2\alpha}$ and for rearranging to obtain a 3TQ with terms in any order
	M1	For a correct attempt using a valid method to solve the 3TQ with two solutions for $\tan\alpha$ Accept use of θ for example in place of α
	A1	For $\tan\alpha = -1 \pm \sqrt{2}$
(ii)(a)	**General principle**. (The second and third M marks can be carried out in either order) First M1 – For expanding the two addition formulae **correctly and** equating. Second M1 – For collecting up like terms to obtain $\sin x(.....) = \cos x(.....)$ allow sign errors. Third M1 – For use of the $\tan = \dfrac{\sin}{\cos}$ identity Final A1 – For the given answer with no errors	

Type MS · Deriving the double-angle formulae

Answer 14 of 24 · Question 7 part a · 2 marks

2022 JUN · Paper 02 · extract 2 of 3 · find part a in this mark scheme extract

	They can work in terms of sin30°, cos30° or tan30° or alternatively $\frac{1}{2}, \frac{\sqrt{3}}{2}, \frac{1}{\sqrt{3}}$ respectively	
	M1	For the correct use of $\cos(A-B)$ and $\sin(A+B)$ to obtain $\cos x\cos 30^\circ + \sin x\sin 30^\circ = \sin x\cos 30^\circ + \cos x\sin 30^\circ$
	M1	For collecting up like terms to obtain: $\sin x(\sin 30^\circ - \cos 30^\circ) = \cos x(\sin 30^\circ - \cos 30^\circ)$ **NB**: This is an A mark in Epen
	M1	For use of $\tan = \frac{\sin}{\cos}$ at any stage in their working.
	A1	For $\tan x = 1$
	ALT	
	M1	For the correct use of $\cos(A-B)$ and $\sin(A+B)$ to obtain $\cos x\cos 30^\circ + \sin x\sin 30^\circ = \sin x\cos 30^\circ + \cos x\sin 30^\circ$
	M1	For dividing through by $\cos x$ and collecting up like terms to obtain: $\frac{\cos x\cos 30^\circ + \sin x\sin 30^\circ}{\cos x} = \frac{\sin x\cos 30^\circ + \cos x\sin 30^\circ}{\cos x}$ $\Rightarrow \frac{\sin x}{\cos x}(\sin 30^\circ - \cos 30^\circ) = \sin 30^\circ - \cos 30^\circ$ **OR** For dividing through by $\cos x$ **and** $\cos 30^\circ$ and also collecting up like terms to obtain: $\frac{\cos x\cos 30^\circ + \sin x\sin 30^\circ}{\cos x\cos 30^\circ} = \frac{\sin x\cos 30^\circ + \cos x\sin 30^\circ}{\cos x\cos 30^\circ}$ $\Rightarrow 1 + \frac{\sin x\sin 30^\circ}{\cos x\cos 30^\circ} = \frac{\sin x}{\cos x} + \frac{\sin 30^\circ}{\cos 30^\circ} \Rightarrow \frac{\sin x}{\cos x}\left(\frac{\sin 30^\circ}{\cos 30^\circ} - 1\right) = \frac{\sin 30^\circ}{\cos 30^\circ} - 1$
	M1	For use of $\tan = \frac{\sin}{\cos}$ at any stage in their working.
	A1	For $\tan x = 1$
(ii)(b)	**M1**	For solving the equation $\tan 2y = 1 \Rightarrow \tan y = -1 \pm \sqrt{2} \Rightarrow y = ..., ...$ (leading to two values for y) **NB**: This is an B mark in Epen
	A1	$y = -67.5^\circ, 22.5^\circ$ Both values required. **NB**: This is an B mark in Epen
	ALT	
	M1	For solving the equation $\tan 2y = 1$ to achieve at least one correct value for $2y$ $2y = \tan^{-1}1 = 45^\circ, -135^\circ$ [Accept even for example 225° for this mark]
	A1	$y = -67.5^\circ, 22.5^\circ$ Both values required. For extra angles in range withhold this mark. Extra angles outside of the range – ignore. **NB**: This is an B mark in Epen

Type MS · Deriving the double-angle formulae

Answer 14 of 24 · Question 7 part a · 2 marks

2022 JUN · Paper 02 · extract 3 of 3 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Deriving the double-angle formulae

Answer 15 of 24 · Question 9 part a · 2 marks

2024 NOV · Paper 02 · extract 1 of 3 · find part a in this mark scheme extract

number		
9 (a)	$\cos 2\theta = \{\cos\theta\cos\theta - \sin\theta\sin\theta\} = \cos^2\theta - \sin^2\theta = \cos^2\theta - (1-\cos^2\theta)$ $= 2\cos^2\theta - 1$ *	M1 A1*cso [2]
(b)	$\int_{\frac{\pi}{3}}^{\frac{3\pi}{4}} \cos 2\theta \{\mathrm{d}\theta\}$ $\left[\frac{\sin 2\theta}{2}\right]_{\frac{\pi}{3}}^{\frac{3\pi}{4}} = \frac{\sin 2\left(\frac{3\pi}{4}\right)}{2} - \frac{\sin 2\left(\frac{\pi}{3}\right)}{2} = -\frac{2+\sqrt{3}}{4}$	M1 M1M1A1 [4]
(c)	$2\cos^2\theta - 1 = -\cos\theta$ $2\cos^2\theta + \cos\theta - 1 = 0 \Rightarrow (2\cos\theta - 1)(\cos\theta + 1) = 0 \Rightarrow \cos\theta = \ldots$ $\theta = \frac{\pi}{3}$ $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} -\cos\theta \, \mathrm{d}\theta$ $= \left[-\sin\theta\right]_{\frac{\pi}{3}}^{\frac{\pi}{2}}$ $= -\sin\frac{\pi}{2} - \left(-\sin\frac{\pi}{3}\right) \left\{= \frac{-2+\sqrt{3}}{2}\right\}$ $\frac{2+\sqrt{3}}{4} - \frac{2-\sqrt{3}}{2}$ $\frac{-2+3\sqrt{3}}{4}$	M1 M1 A1 M1 M1 M1 ddM1 A1 [8]
ALT for last 5 marks	$\int_{-\frac{\pi}{3}}^{\frac{\pi}{2}} (-\cos\theta - \cos 2\theta)\{d\theta\} \{+\} \int_{\frac{\pi}{2}}^{\frac{3\pi}{4}} (-\cos 2\theta)\{d\theta\}$ $= \left[-\sin\theta - \frac{\sin 2\theta}{2}\right]_{-\frac{\pi}{3}}^{\frac{\pi}{2}} \{+\} \left[-\frac{\sin 2\theta}{2}\right]_{\frac{\pi}{2}}^{\frac{3\pi}{4}}$ $= \left(-\sin\frac{\pi}{2} - \frac{\sin 2\times\frac{\pi}{2}}{2}\right) - \left(-\sin\frac{\pi}{3} - \frac{\sin 2\times\frac{\pi}{3}}{2}\right) \{+\} \left(-\frac{\sin 2\times\frac{3\pi}{4}}{2}\right) - \left(-\frac{\sin 2\times\frac{\pi}{2}}{2}\right)$ $= \frac{-4+3\sqrt{3}}{4} + \frac{1}{2}$ $= \frac{-2+3\sqrt{3}}{4}$	M1 M1 M1 ddM1 A1
		Total 14 marks

Type MS · Deriving the double-angle formulae

Answer 15 of 24 · Question 9 part a · 2 marks

2024 NOV · Paper 02 · extract 2 of 3 · find part a in this mark scheme extract

Part	Mark	Additional Guidance
(a)	M1	For $\cos^2\theta-\sin^2\theta=\cos^2\theta-\left(1-\cos^2\theta\right)$ or $\cos^2\theta-\sin^2\theta=\cos^2\theta+\cos^2\theta-1$
	A1*cso	Obtains the correct result with no errors seen. Must have $\cos 2\theta$ on the lhs in their working, does not have to be in the final line provided there is a chain of reasoning.
(b)	M1	For replacement of $2\cos^2\theta-1$ with $\cos 2\theta$,ignore limits
	M1	For an attempt to integrate to an expression of the form $\pm\frac{\sin 2\theta}{2}$
	M1	Correct substitution of the **given** limits to a **changed** expression in terms of $\sin 2\theta$ and subtracts the correct way around. A correct value from a correctly integrated expression scores this mark, no need to see explicit substitution
	A1	Correct answer. isw
(c)	M1	Correctly equates the two curves.
	M1	Rearranges their 3TQ = 0 and attempts to solve (see general guidance). Can be implied by their correct $\cos\theta$ values, $\frac{1}{2}$ **and** –1 from a correct 3TQ, if their 3TQ is not correct, a clear method must be shown.
	A1	Correct angle found. Ignore extra angles.
	M1	**Main**: For stating the correct integral with "correct" limits of their $\frac{\pi}{3}$ and $\frac{\pi}{2}$ $\pm\int_{\frac{\pi}{3}}^{\frac{\pi}{2}}-\cos\theta\,\mathrm{d}\theta$ and $\pm\int_{\frac{\pi}{3}}^{\frac{3\pi}{4}}(2\cos^2\theta-1)\mathrm{d}\theta$ (the later integral can be implied by later work) (their $\frac{\pi}{3}$ must be positive and < $\frac{\pi}{2}$) **ALT**: Correct integrals $\pm\int_{\frac{\pi}{3}}^{\frac{\pi}{2}}(-\cos\theta-\cos 2\theta)\{\mathrm{d}\theta\}$ and $\pm\int_{\frac{\pi}{2}}^{\frac{3\pi}{4}}(-\cos 2\theta)\{\mathrm{d}\theta\}$ oe
	M1 **A1 on Epen**	For a correct integration of **all** their integrals in $\cos 2\theta$ and/or $\cos\theta$ only. Ignore limits
	M1	For correct **explicit** substitution of their limits into **either changed** expression, subtracts the correct way around – substitution of 0 must be seen unless substitution into their changed function gives 0 Can be implied by $\pm\left(\frac{-2+3\sqrt{3}}{4}\right)$ from correct working
	ddM1	Correct strategy finding the required area: **Main**: For subtracting the modulus of their 2 integrals. oe. Either way around. **ALT**: Add the values of two positive area "$\frac{-4+3\sqrt{3}}{4}$" and "$\frac{1}{2}$" oe Dependent on previous **three** method marks
	A1	Correct answer, simplest form and exact area of R

Type MS · Deriving the double-angle formulae

Answer 15 of 24 · Question 9 part a · 2 marks

2024 NOV · Paper 02 · extract 3 of 3 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Deriving the double-angle formulae

Answer 16 of 24 · Question 11 part a · 3 marks

2023 JUN · Paper 02 (regional R) · extract 1 of 4 · find part a in this mark scheme extract

11(a)	$\cos 2A = \cos^2 A - \sin^2 A \Rightarrow \cos 2A = (1-\sin^2 A) - \sin^2 A = 1 - 2\sin^2 A$ $2\sin^2 A = 1 - \cos 2A \Rightarrow \sin^2 A = \frac{1}{2}(1-\cos 2A)$ *	M1M1 A1cso [3]
(b)	$\cos^2 A = \frac{1}{2}(1+\cos 2A)$ $\sin^4 x + \cos^4 x = \left(\left(\frac{1-\cos 2x}{2}\right)^2 + \left(\frac{1+\cos 2x}{2}\right)^2\right)$ $= \frac{1}{4}\left((1-\cos 2x)^2 + (1+\cos 2x)^2\right)$ $= \frac{1}{4}\left(1 - 2\cos 2x + \cos^2 2x + 1 + 2\cos 2x + \cos^2 2x\right)$ $= \frac{1}{4}\left(2 + 2\cos^2 2x\right) = \left\{\frac{1}{2}\left(1+\cos^2 2x\right)\right\}$ $\left\{\cos^2 2x = \frac{1+\cos 4x}{2}\right\}$ $\sin^4 x + \cos^4 x = \frac{1}{2}\left(1 + \frac{1+\cos 4x}{2}\right) = \frac{3+\cos 4x}{4}$ *	B1 M1 A1 M1A1 cso [5]
(c)	$5\sin 2\theta + 6 = 8\left(\frac{3+\cos 2\theta}{4}\right) \Rightarrow 5\sin 2\theta + 6 = 6 + 2\cos 2\theta$ $\Rightarrow 5\sin 2\theta = 2\cos 2\theta \Rightarrow \frac{\sin 2\theta}{\cos 2\theta} = \frac{2}{5} \Rightarrow \tan 2\theta = \frac{2}{5}$ $2\theta = 21.801...^\circ, 201.801...^\circ, 381.801...^\circ$ $\Rightarrow \theta = 10.9^\circ, 100.9^\circ$ Penalise extra angles in range by withholding the final A mark. Extra angles out of range – ignore.	M1 M1 A1 A1 [4]
		Total 12 marks

Part	Mark	Notes
(a)	M1	For using the summation formula for cos 2A They must start with either $\cos 2A = \cos^2 A - \sin^2 A$ or $\cos(A+A) = \cos A \cos A - \sin A \sin A$
	M1	For eliminating $\cos^2 A$ using $\cos^2 A + \sin^2 A = 1$ and attempting to rearrange to the required result. This is not dependent on the first M mark, so if they start with another identity for cos2A they can still get this mark.
	A1 cso	For the correct identity with no errors seen. This is a given result.
	NB Some candidates work backwards – that is fine, please follow their working.	

Type MS · Deriving the double-angle formulae

Answer 16 of 24 · Question 11 part a · 3 marks

2023 JUN · Paper 02 (regional R) · extract 2 of 4 · find part a in this mark scheme extract

Working with a different variable. If they work in this part with a different variable (eg A) then award all the marks as appropriate up to the last mark. If they leave their final answer in terms of another variable, withhold the final A mark only. If they however, change to x on the final line award all the marks [provided everything is correct]		
(b)	**Main method**	
	B1	For use of the correct identity for $\cos^2 x$ $\left[\cos^2 x = \frac{1}{2}(1+\cos 2x)\right]$ or if they convert $\cos^2 x$ to $\sin^2 x$ use the Pythagorean identity $\cos^2 x = 1-\sin^2 x$ and apply the given identity.
	M1	For squaring both identities. This must be a correct expansion. Ft their identity for $\left[\cos^2 x = \frac{1}{2}(1+\cos 2x)\right]$
	A1	For collecting like terms and obtaining $\frac{1}{2}(1+\cos^2 2x)$ oe. For example, $\frac{1}{4}(2+2\cos^2 2x)$
	M1	For applying the identity: $\cos^2 A = \frac{1}{2}(1+\cos 2A) \Rightarrow \left[\cos^2 2A = \frac{1}{2}(1+\cos 4A)\right]$ on $\cos^2 2x$ **only** again to achieve an expression in $\cos 4x$ **only**
	A1 cso	For the correct identity with no errors seen. This is a given result. You must check every line of their work carefully.
	ALT 1	
	B1	For use of the correct identity for $\sin 2A = 2\sin A\cos A$ [seen later in their working].
	M1	For using the expansion of $\sin^2 x+\cos^2 x$ as follows $\sin^4 x+\cos^4 x = (\sin^2 x+\cos^2 x)^2 - 2\sin^2 x\cos^2 x$ This must be correct
	A1	For obtaining $\sin^4 x+\cos^4 x = 1-\frac{1}{2}\sin^2 2x$ oe. For example; $\frac{1}{2}(2-\sin^2 2x)$
	M1	For applying the given identity on $\sin^2 2x$ **only** to achieve an expression in $\cos 4x$ **only** $\sin^2 A = \frac{1}{2}(1-\cos 2A) \Rightarrow \left[\sin^2 2A = \frac{1}{2}(1-\cos 4A)\right]$ $\sin^4 x+\cos^4 x = 1-\frac{1}{2}\left(\frac{1-\cos 4x}{2}\right) \Rightarrow \left(\sin^4 x+\cos^4 x = \frac{3+\cos 4x}{4}\right)$
	A1 cso	For the correct identity with no errors seen. This is a given result. You must check every line of their work carefully.

Type MS · Deriving the double-angle formulae

Answer 16 of 24 · Question 11 part a · 3 marks

2023 JUN · Paper 02 (regional R) · extract 3 of 4 · find part a in this mark scheme extract

	ALT 2 – Works backwards from the given result	
	B1	For use of the correct identity for $\sin 2A = 2\sin A\cos A$ [seen later in their working].
	M1	Applies the cos2A identity and converts 3 into $3\left(\sin^2 2x+\cos^2 2x\right)$ $\frac{3+\cos 4x}{4}=\frac{3\sin^2 2x+3\cos^2 2x+\left(\cos^2 2x-\sin^2 2x\right)}{4}$
	A1	Obtains $\frac{3+\cos 4x}{4}=\frac{4\cos^2 2x+2\sin^2 2x}{4}$
	M1	Applies the cos2A identity and expands the bracket. $\frac{3+\cos 4x}{4}=\frac{4\left(\cos^2 x-\sin^2 x\right)^2+8\sin^2 x\cos^2 x}{4}$ $=\frac{4\cos^4 x-8\sin^2 x\cos^2 x+4\sin^4 x+8\sin^2 x\cos^2 x}{4}$
	A1 cso	Simplifies to the required result with no errors seen $\frac{3+\cos 4x}{4}=\sin^4 x+\cos^4 x$ *

(c)	M1	For obtaining the correct equation in terms of $\sin 2\theta$ and $\cos 2\theta$ $6+2\cos 2\theta=5\sin 2\theta+6$ Accept unsimplified, accept even: $8\left(\frac{3+\cos 2\theta}{4}\right)=5\sin 2\theta+6$
	M1	For using the $\tan 2\theta=\frac{\sin 2\theta}{\cos 2\theta}$ identity correctly on their expression following an expression in the form $A\cos 2\theta=B\sin 2\theta$ and must be in terms of 2θ
	A1	For achieving at least one correct angle for 2θ **NB This is an M mark in Epen**
	A1	For awrt both 10.9° and 100.9° Penalise extra angles in range by withholding the final A mark. Extra angles out of range – ignore.

Type MS · Deriving the double-angle formulae

Answer 16 of 24 · Question 11 part a · 3 marks

2023 JUN · Paper 02 (regional R) · extract 4 of 4 · find part a in this mark scheme extract

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Type MS · Deriving the double-angle formulae

Answer 17 of 24 · Question 10 part b.i · 0 marks

2022 JAN · Paper 01 (regional R) · extract 1 of 2 · find part b.i in this mark scheme extract

10 (a)	$f\left(\frac{3}{4}\right)=16\left(\frac{3}{4}\right)^3+11\left(\frac{3}{4}\right)-15$ * $\left(\frac{27}{4}+\frac{33}{4}-15\right)=0$	M1 A1cso [2]
(b)(i)	$[\sin 2\theta=\sin(\theta+\theta)]=\sin\theta\cos\theta+\sin\theta\cos\theta$	M1
	$(\sin 2\theta=)2\sin\theta\cos\theta$ *	A1cso [2]
(b)(ii)	$[\cos 2\theta=\cos(\theta+\theta)=]\cos\theta\cos\theta-\sin\theta\sin\theta=\cos^2\theta-\sin^2\theta$	M1
	$=\cos^2\theta-(1-\cos^2\theta)=2\cos^2\theta-1$ *	M1 A1cso [3]
(c)	$27\cos\theta(2\cos^2\theta-1)+19\sin\theta(2\sin\theta\cos\theta)-15(=0)$	M1
	$54\cos^3\theta-27\cos\theta+38(1-\cos^2\theta)\cos\theta-15(=0)$	M1
	$54\cos^3\theta-27\cos\theta+38\cos\theta-38\cos^3\theta-15(=0)$ $(16\cos^3\theta+11\cos\theta-15=0)$ oe	A1
	(When $x=\cos\theta$) $16x^3+11x-15=0$ *	A1cso [4]
(d)	$(16x^3+11x-15=0\Rightarrow(4x-3)(4x^2+3x+5)=0)\rightarrow 4x^2+3x+5$	M1
	$b^2-4ac=9-80<0$ (So no real roots)	M1 A1
	(Only solution is) $4x-3=0$ or $4\cos\theta-3=0$ so $\cos\theta=\frac{3}{4}$ *	A1cso [4]
		Total 15 marks

Type MS · Deriving the double-angle formulae

Answer 17 of 24 · Question 10 part b.i · 0 marks

2022 JAN · Paper 01 (regional R) · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Scheme	Marks

Type MS · Deriving the double-angle formulae

Answer 18 of 24 · Question 10 part b.ii · 5 marks

2022 JAN · Paper 01 (regional R) · extract 1 of 2 · find part b.ii in this mark scheme extract

10 (a)	$f\left(\frac{3}{4}\right)=16\left(\frac{3}{4}\right)^3+11\left(\frac{3}{4}\right)-15$ * $\left(\frac{27}{4}+\frac{33}{4}-15\right)=0$	M1 A1cso [2]
(b)(i)	$[\sin 2\theta=\sin(\theta+\theta)]=\sin\theta\cos\theta+\sin\theta\cos\theta$	M1
	$(\sin 2\theta=)2\sin\theta\cos\theta$ *	A1cso [2]
(b)(ii)	$[\cos 2\theta=\cos(\theta+\theta)=]\cos\theta\cos\theta-\sin\theta\sin\theta=\cos^2\theta-\sin^2\theta$	M1
	$=\cos^2\theta-(1-\cos^2\theta)=2\cos^2\theta-1$ *	M1 A1cso [3]
(c)	$27\cos\theta(2\cos^2\theta-1)+19\sin\theta(2\sin\theta\cos\theta)-15(=0)$	M1
	$54\cos^3\theta-27\cos\theta+38(1-\cos^2\theta)\cos\theta-15(=0)$	M1
	$54\cos^3\theta-27\cos\theta+38\cos\theta-38\cos^3\theta-15(=0)$ $(16\cos^3\theta+11\cos\theta-15=0)$ oe	A1
	(When $x=\cos\theta$) $16x^3+11x-15=0$ *	A1cso [4]
(d)	$(16x^3+11x-15=0\Rightarrow(4x-3)(4x^2+3x+5)=0)\rightarrow 4x^2+3x+5$	M1
	$b^2-4ac=9-80<0$ (So no real roots)	M1 A1
	(Only solution is) $4x-3=0$ or $4\cos\theta-3=0$ so $\cos\theta=\frac{3}{4}$ *	A1cso [4]
		Total 15 marks

Type MS · Deriving the double-angle formulae

Answer 18 of 24 · Question 10 part b.ii · 5 marks

2022 JAN · Paper 01 (regional R) · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Scheme	Marks

Type MS · Deriving the double-angle formulae

Answer 19 of 24 · Question 10 part a.i · 0 marks

2023 NOV · Paper 01 · extract 1 of 5 · find part a.i in this mark scheme extract

Question	Scheme	Marks
10(a)(i) (ii)	$\sin(A+A)=\sin A\cos A+\cos A\sin A=2\sin A\cos A$ *	B1 cso
	$\cos(A+A)=\cos A\cos A-\sin A\sin A$	
	$\Rightarrow \cos 2A=\cos^2 A-\sin^2 A$	M1
	$=\cos^2 A-(1-\cos^2 A)$	A1*cso
	$=2\cos^2 A-1$ *	[3]
(b)	$\left(\frac{2\tan\theta}{1+\tan^2\theta}=\right)\frac{2\frac{\sin\theta}{\cos\theta}}{1+\frac{\sin^2\theta}{\cos^2\theta}}$	M1
	$=\frac{2\frac{\sin\theta}{\cos\theta}}{\frac{\cos^2\theta+\sin^2\theta}{\cos^2\theta}}$ or $\frac{2\frac{\sin\theta}{\cos\theta}}{1+\frac{1}{\cos^2\theta}-\frac{\cos^2\theta}{\cos^2\theta}}$ or $\frac{2\frac{\sin\theta}{\cos\theta}}{1+\frac{\sin^2\theta}{\cos^2\theta}}\times\frac{\cos^2\theta}{\cos^2\theta}$ or $\frac{2\sin\theta\cos\theta}{\cos^2\theta+\sin^2\theta}$	dM1
	$=\frac{2\frac{\sin\theta}{\cos\theta}}{\frac{1}{\cos^2\theta}}\left(=2\frac{\sin\theta}{\cos\theta}\times\cos^2\theta\right)$ or $\left(\frac{2\sin\theta\cos\theta}{\cos^2\theta+\sin^2\theta}\right)=2\sin\theta\cos\theta$	ddM1
	$=\sin 2\theta$ *	A1*cso [4]
ALT	$\left(\frac{2\tan\theta}{1+\tan^2\theta}=\right)\frac{2\frac{\sin\theta}{\cos\theta}}{1+\frac{\sin^2\theta}{\cos^2\theta}}$	M1
	$=\frac{2\frac{\sin\theta}{\cos\theta}}{1+\tan^2\theta}$	dM1
	$\cos^2\theta+\sin^2\theta=1\Rightarrow\tan^2\theta+1=\frac{1}{\cos^2\theta}\Rightarrow\frac{2\frac{\sin\theta}{\cos\theta}}{\frac{1}{\cos^2\theta}}\left(=2\frac{\sin\theta}{\cos\theta}\times\cos^2\theta\right)=2\sin\theta\cos\theta$	ddM1
	$=\sin 2\theta$ *	A1*cso [4]
(c)	$\frac{5\tan\left(x+\frac{\pi}{6}\right)}{1+\tan^2\left(x+\frac{\pi}{6}\right)}=\left(1-2\cos^2\left(x+\frac{\pi}{6}\right)\right)\Rightarrow\frac{5}{2}\sin\left(2\left(x+\frac{\pi}{6}\right)\right)=\left(1-2\cos^2\left(x+\frac{\pi}{6}\right)\right)$	M1
	$1-2\cos^2\left(x+\frac{\pi}{6}\right)=-\cos\left(2x+\frac{\pi}{3}\right)$	M1
	$\left(\frac{5}{2}\sin\left(2x+\frac{\pi}{3}\right)=-\cos\left(2x+\frac{\pi}{3}\right)\Rightarrow\right)\tan\left(2x+\frac{\pi}{3}\right)=-\frac{2}{5}$	M1
	$\left(\Rightarrow 2x+\frac{\pi}{3}=\right)-0.38050,\quad 2.76108$	M1
	$(\Rightarrow x=)-0.714,\quad 0.857$	A1A1 [6]

Type MS · Deriving the double-angle formulae

Answer 19 of 24 · Question 10 part a.i · 0 marks

2023 NOV · Paper 01 · extract 2 of 5 · find part a.i in this mark scheme extract

(d)	$I = \int_0^{\frac{\pi}{2}} (2\sin 2\theta - \cos 5\theta + 2)\ \{\mathrm{d}\theta\}$	M1
	$= \left[\frac{-2\cos 2\theta}{2} - \frac{\sin 5\theta}{5} + 2\theta\right]_0^{\frac{\pi}{2}}$	M1
	$= \left(-\cos 2\left(\frac{\pi}{2}\right) - \frac{\sin 5\left(\frac{\pi}{2}\right)}{5} + 2\times\left(\frac{\pi}{2}\right)\right) - \left(-\cos 2(0) - \frac{\sin 5(0)}{5} + 2\times 0\right)$	M1
	$\left(= \left(1 - \frac{1}{5} + \pi\right) - (-1 - 0 + 0)\right) = \frac{9}{5} + \pi$	A1 [4]
		Total 17 marks

Type MS · Deriving the double-angle formulae

Answer 19 of 24 · Question 10 part a.i · 0 marks

2023 NOV · Paper 01 · extract 3 of 5 · find part a.i in this mark scheme extract

Part	Mark	Notes
(a)	B1 cso	For $\sin(A+A)$ or $\sin 2A = \sin A\cos A + \cos A\sin A = 2\sin A\cos A$ * No steps missing or incorrect work. Condone $\sin A + A$ with brackets missing if all other work correct.
	M1	For $\cos A\cos A - \sin A\sin A \Rightarrow \cos^2 A - \sin^2 A$
	A1* cso	For $\cos(A+A)$ or $\cos 2A = \cos A\cos A - \sin A\sin A \Rightarrow \cos^2 A - \sin^2 A = \cos^2 A - (1-\cos^2 A)$ $= 2\cos^2 A - 1$ No steps missing, no incorrect work. Condone $\cos A + A$ with brackets missing if all other work correct.
Students who miss 'A' in their working, (eg coscos – sinsin) may be awarded M mark, but not A or B marks.		
(b)	M1	Uses $\tan\theta = \frac{\sin\theta}{\cos\theta}$ and $\tan^2\theta = \frac{\sin^2\theta}{\cos^2\theta}$ in the given identity. Allow the 2 to be missing for this mark **but not the final A mark.**
	dM1	For $1 + \frac{\sin^2\theta}{\cos^2\theta} = \frac{\cos^2\theta + \sin^2\theta}{\cos^2\theta}$ or multiplying by $\frac{\cos^2\theta}{\cos^2\theta}$ to give $\left(\frac{2\frac{\sin\theta}{\cos\theta}}{1+\frac{\sin^2\theta}{\cos^2\theta}} = \right) \frac{2\sin\theta\cos\theta}{\cos^2\theta+\sin^2\theta}$ The $\times\frac{\cos^2\theta}{\cos^2\theta}$ does not need to be explicitly written. Dependent on previous method mark.
	ddM1	For minimum steps (and dependent on previous method marks): $\frac{2\frac{\sin\theta}{\cos\theta}}{\frac{1}{\cos^2\theta}}$ or $\frac{2\sin\theta\cos\theta}{\cos^2\theta+\sin^2\theta} = 2\sin\theta\cos\theta$
	A1*cso	Achieves the required result – minimum steps as shown, no errors. **Steps in brackets do not need to be shown.**
ALT	M1	Uses $\tan\theta = \frac{\sin\theta}{\cos\theta}$ and $\tan^2\theta = \frac{\sin^2\theta}{\cos^2\theta}$ in the given identity.
	dM1	For replacing $1 + \frac{\sin^2\theta}{\cos^2\theta}$ with $1 + \tan^2\theta$ Dependent on previous method mark.
	ddM1	For replacing $\tan^2\theta + 1 = \frac{1}{\cos^2\theta} \Rightarrow \frac{2\frac{\sin\theta}{\cos\theta}}{\frac{1}{\cos^2\theta}} = 2\sin\theta\cos\theta$ Dependent on both previous method marks.
	A1*cso	Achieves the required result – minimum steps as shown, no errors. **Steps in brackets do not need to be shown.**
Students who miss θ in their working, may be awarded M marks, but not the A mark.		

Type MS · Deriving the double-angle formulae

Answer 19 of 24 · Question 10 part a.i · 0 marks

2023 NOV · Paper 01 · extract 4 of 5 · find part a.i in this mark scheme extract

Allow work throughout where candidates have let, for example $\theta = x + \frac{\pi}{6}$ and worked in θ and 2θ throughout in (c)		
(c)	M1	Rearranges the given equation into $\frac{5\tan\left(x+\frac{\pi}{6}\right)}{1+\tan^2\left(x+\frac{\pi}{6}\right)} = \left(1-2\cos^2\left(x+\frac{\pi}{6}\right)\right)$ **and** changes the left hand side into $P\sin\left(2\left(x+\frac{\pi}{6}\right)\right)$ where P is any constant.
	M1	Correctly states or uses $1-2\cos^2\left(x+\frac{\pi}{6}\right) = -\cos\left(2x+\frac{\pi}{3}\right)$ (condone being replaced with $-\cos 2A$)
	M1	For $\tan\left(2x+\frac{\pi}{3}\right) = Q$ where Q is any constant.
	M1	For -0.38050 or 2.76108 This may be awarded for either value being seen, ignore any extra values seen. Accept any values with at least 3 significant figures.
	A1	For $(x=)-0.714$ or $(x=)0.857$ This may be awarded for either value being seen, ignore any extra values seen. Accept any value with more than 3 significant rounding to those given.
	A1	For $(x=)-0.714$ and $(x=)0.857$ Ignore any extra values **out of range**. Extra values in range will mean A0. Accept any value with more than 3 significant rounding to those given.
Students must use θ or $x+\frac{\pi}{6}$ and 2θ or $2\left(x+\frac{\pi}{6}\right)$		
(d)	M1	For changing the integral into $(I=)\int_0^{\frac{\pi}{2}}(2\sin 2\theta - \cos 5\theta + 2)\,\{\mathrm{d}\theta\}$ $\mathrm{d}\theta$ may be omitted. If the integral sign and limits are not present, these can be implied by a clear attempt in later work to integrate all the terms and a substitution of limits. Students must use 2θ and 5θ in their working.
	M1	For a minimum of 2 out of the 3 terms shown above correctly integrated. Students must use 2θ and 5θ in their working.
	M1	For substituting in the given limits the correct way around into their changed function with at least two terms, there must be at least one explicit substitution of each limit. If the candidate has a constant term as part of the function, they can omit this from this step (as the terms cancel out anyway). This mark may be implied by a correct answer. If the candidate is working with an incorrect function and substitution of 0 clearly gives 0, this substitution does not need to be shown.
	A1	For $\frac{9}{5}+\pi$ The approximate decimal value 4.941…will **usually** indicate correct working (as a guide), but a substitution of limits will be needed to get the 3rd method mark.
In this question, students are directed to use calculus, so must show a correct integration. The penultimate method mark may be implied from a fully correct answer, but if the answer is not correct, we must see substitution of limits as stated.		

Type MS · Deriving the double-angle formulae

Answer 19 of 24 · Question 10 part a.i · 0 marks

2023 NOV · Paper 01 · extract 5 of 5 · find part a.i in this mark scheme extract

Question	Scheme	Marks
	As with many log questions, there are a number of ways of reaching a correct answer. **The first scheme shows 3 common routes, with a set of general principles in the notes.** **Not all variations in methods will have been captured, please mark to general principles, referring ANY work where this is not possible and worthy of marks to REVIEW.** **This includes work that converts everything to completely new bases.**	

Type MS · Deriving the double-angle formulae

Answer 20 of 24 · Question 10 part a.ii · 3 marks

2023 NOV · Paper 01 · extract 1 of 5 · find part a.ii in this mark scheme extract

Question	Scheme	Marks
10(a)(i) (ii)	$\sin(A+A)=\sin A\cos A+\cos A\sin A=2\sin A\cos A$ *	B1 cso
	$\cos(A+A)=\cos A\cos A-\sin A\sin A$	
	$\Rightarrow \cos 2A=\cos^2 A-\sin^2 A$	M1
	$=\cos^2 A-(1-\cos^2 A)$	A1*cso
	$=2\cos^2 A-1$ *	[3]
(b)	$\left(\frac{2\tan\theta}{1+\tan^2\theta}=\right)\frac{2\frac{\sin\theta}{\cos\theta}}{1+\frac{\sin^2\theta}{\cos^2\theta}}$	M1
	$=\frac{2\frac{\sin\theta}{\cos\theta}}{\frac{\cos^2\theta+\sin^2\theta}{\cos^2\theta}}$ or $\frac{2\frac{\sin\theta}{\cos\theta}}{1+\frac{1}{\cos^2\theta}-\frac{\cos^2\theta}{\cos^2\theta}}$ or $\frac{2\frac{\sin\theta}{\cos\theta}}{1+\frac{\sin^2\theta}{\cos^2\theta}}\times\frac{\cos^2\theta}{\cos^2\theta}$ or $\frac{2\sin\theta\cos\theta}{\cos^2\theta+\sin^2\theta}$	dM1
	$=\frac{2\frac{\sin\theta}{\cos\theta}}{\frac{1}{\cos^2\theta}}\left(=2\frac{\sin\theta}{\cos\theta}\times\cos^2\theta\right)$ or $\left(\frac{2\sin\theta\cos\theta}{\cos^2\theta+\sin^2\theta}\right)=2\sin\theta\cos\theta$	ddM1
	$=\sin 2\theta$ *	A1*cso [4]
ALT	$\left(\frac{2\tan\theta}{1+\tan^2\theta}=\right)\frac{2\frac{\sin\theta}{\cos\theta}}{1+\frac{\sin^2\theta}{\cos^2\theta}}$	M1
	$=\frac{2\frac{\sin\theta}{\cos\theta}}{1+\tan^2\theta}$	dM1
	$\cos^2\theta+\sin^2\theta=1\Rightarrow\tan^2\theta+1=\frac{1}{\cos^2\theta}\Rightarrow\frac{2\frac{\sin\theta}{\cos\theta}}{\frac{1}{\cos^2\theta}}\left(=2\frac{\sin\theta}{\cos\theta}\times\cos^2\theta\right)=2\sin\theta\cos\theta$	ddM1
	$=\sin 2\theta$ *	A1*cso [4]
(c)	$\frac{5\tan\left(x+\frac{\pi}{6}\right)}{1+\tan^2\left(x+\frac{\pi}{6}\right)}=\left(1-2\cos^2\left(x+\frac{\pi}{6}\right)\right)\Rightarrow\frac{5}{2}\sin\left(2\left(x+\frac{\pi}{6}\right)\right)=\left(1-2\cos^2\left(x+\frac{\pi}{6}\right)\right)$	M1
	$1-2\cos^2\left(x+\frac{\pi}{6}\right)=-\cos\left(2x+\frac{\pi}{3}\right)$	M1
	$\left(\frac{5}{2}\sin\left(2x+\frac{\pi}{3}\right)=-\cos\left(2x+\frac{\pi}{3}\right)\Rightarrow\right)\tan\left(2x+\frac{\pi}{3}\right)=-\frac{2}{5}$	M1
	$\left(\Rightarrow 2x+\frac{\pi}{3}=\right)-0.38050,\quad 2.76108$	M1
	$(\Rightarrow x=)-0.714,\quad 0.857$	A1A1 [6]

Type MS · Deriving the double-angle formulae

Answer 20 of 24 · Question 10 part a.ii · 3 marks

2023 NOV · Paper 01 · extract 2 of 5 · find part a.ii in this mark scheme extract

(d)	$I = \int_0^{\frac{\pi}{2}} (2\sin 2\theta - \cos 5\theta + 2)\ \{\mathrm{d}\theta\}$	M1
	$= \left[\frac{-2\cos 2\theta}{2} - \frac{\sin 5\theta}{5} + 2\theta\right]_0^{\frac{\pi}{2}}$	M1
	$= \left(-\cos 2\left(\frac{\pi}{2}\right) - \frac{\sin 5\left(\frac{\pi}{2}\right)}{5} + 2\times\left(\frac{\pi}{2}\right)\right) - \left(-\cos 2(0) - \frac{\sin 5(0)}{5} + 2\times 0\right)$	M1
	$\left(= \left(1 - \frac{1}{5} + \pi\right) - (-1 - 0 + 0)\right) = \frac{9}{5} + \pi$	A1 [4]
		Total 17 marks

Type MS · Deriving the double-angle formulae

Answer 20 of 24 · Question 10 part a.ii · 3 marks

2023 NOV · Paper 01 · extract 3 of 5 · find part a.ii in this mark scheme extract

Part	Mark	Notes
(a)	B1 cso	For $\sin(A+A)$ or $\sin 2A = \sin A\cos A + \cos A\sin A = 2\sin A\cos A$ * No steps missing or incorrect work. Condone $\sin A + A$ with brackets missing if all other work correct.
	M1	For $\cos A\cos A - \sin A\sin A \Rightarrow \cos^2 A - \sin^2 A$
	A1* cso	For $\cos(A+A)$ or $\cos 2A = \cos A\cos A - \sin A\sin A \Rightarrow \cos^2 A - \sin^2 A = \cos^2 A - (1-\cos^2 A)$ $= 2\cos^2 A - 1$ No steps missing, no incorrect work. Condone $\cos A + A$ with brackets missing if all other work correct.
Students who miss 'A' in their working, (eg coscos – sinsin) may be awarded M mark, but not A or B marks.		
(b)	M1	Uses $\tan\theta = \frac{\sin\theta}{\cos\theta}$ and $\tan^2\theta = \frac{\sin^2\theta}{\cos^2\theta}$ in the given identity. Allow the 2 to be missing for this mark **but not the final A mark.**
	dM1	For $1 + \frac{\sin^2\theta}{\cos^2\theta} = \frac{\cos^2\theta + \sin^2\theta}{\cos^2\theta}$ or multiplying by $\frac{\cos^2\theta}{\cos^2\theta}$ to give $\left(\frac{2\frac{\sin\theta}{\cos\theta}}{1+\frac{\sin^2\theta}{\cos^2\theta}} = \right) \frac{2\sin\theta\cos\theta}{\cos^2\theta+\sin^2\theta}$ The $\times\frac{\cos^2\theta}{\cos^2\theta}$ does not need to be explicitly written. Dependent on previous method mark.
	ddM1	For minimum steps (and dependent on previous method marks): $\frac{2\frac{\sin\theta}{\cos\theta}}{\frac{1}{\cos^2\theta}}$ or $\frac{2\sin\theta\cos\theta}{\cos^2\theta+\sin^2\theta} = 2\sin\theta\cos\theta$
	A1*cso	Achieves the required result – minimum steps as shown, no errors. **Steps in brackets do not need to be shown.**
ALT	M1	Uses $\tan\theta = \frac{\sin\theta}{\cos\theta}$ and $\tan^2\theta = \frac{\sin^2\theta}{\cos^2\theta}$ in the given identity.
	dM1	For replacing $1 + \frac{\sin^2\theta}{\cos^2\theta}$ with $1 + \tan^2\theta$ Dependent on previous method mark.
	ddM1	For replacing $\tan^2\theta + 1 = \frac{1}{\cos^2\theta} \Rightarrow \frac{2\frac{\sin\theta}{\cos\theta}}{\frac{1}{\cos^2\theta}} = 2\sin\theta\cos\theta$ Dependent on both previous method marks.
	A1*cso	Achieves the required result – minimum steps as shown, no errors. **Steps in brackets do not need to be shown.**
Students who miss θ in their working, may be awarded M marks, but not the A mark.		

Type MS · Deriving the double-angle formulae

Answer 20 of 24 · Question 10 part a.ii · 3 marks

2023 NOV · Paper 01 · extract 4 of 5 · find part a.ii in this mark scheme extract

Allow work throughout where candidates have let, for example $\theta = x + \frac{\pi}{6}$ and worked in θ and 2θ throughout in (c)		
(c)	M1	Rearranges the given equation into $\frac{5\tan\left(x+\frac{\pi}{6}\right)}{1+\tan^2\left(x+\frac{\pi}{6}\right)} = \left(1-2\cos^2\left(x+\frac{\pi}{6}\right)\right)$ **and** changes the left hand side into $P\sin\left(2\left(x+\frac{\pi}{6}\right)\right)$ where P is any constant.
	M1	Correctly states or uses $1-2\cos^2\left(x+\frac{\pi}{6}\right) = -\cos\left(2x+\frac{\pi}{3}\right)$ (condone being replaced with $-\cos 2A$)
	M1	For $\tan\left(2x+\frac{\pi}{3}\right) = Q$ where Q is any constant.
	M1	For -0.38050 or 2.76108 This may be awarded for either value being seen, ignore any extra values seen. Accept any values with at least 3 significant figures.
	A1	For $(x=)-0.714$ or $(x=)0.857$ This may be awarded for either value being seen, ignore any extra values seen. Accept any value with more than 3 significant rounding to those given.
	A1	For $(x=)-0.714$ and $(x=)0.857$ Ignore any extra values **out of range**. Extra values in range will mean A0. Accept any value with more than 3 significant rounding to those given.
Students must use θ or $x+\frac{\pi}{6}$ and 2θ or $2\left(x+\frac{\pi}{6}\right)$		
(d)	M1	For changing the integral into $(I=)\int_0^{\frac{\pi}{2}}(2\sin 2\theta - \cos 5\theta + 2)\,\{\mathrm{d}\theta\}$ $\mathrm{d}\theta$ may be omitted. If the integral sign and limits are not present, these can be implied by a clear attempt in later work to integrate all the terms and a substitution of limits. Students must use 2θ and 5θ in their working.
	M1	For a minimum of 2 out of the 3 terms shown above correctly integrated. Students must use 2θ and 5θ in their working.
	M1	For substituting in the given limits the correct way around into their changed function with at least two terms, there must be at least one explicit substitution of each limit. If the candidate has a constant term as part of the function, they can omit this from this step (as the terms cancel out anyway). This mark may be implied by a correct answer. If the candidate is working with an incorrect function and substitution of 0 clearly gives 0, this substitution does not need to be shown.
	A1	For $\frac{9}{5}+\pi$ The approximate decimal value 4.941…will **usually** indicate correct working (as a guide), but a substitution of limits will be needed to get the 3rd method mark.
In this question, students are directed to use calculus, so must show a correct integration. The penultimate method mark may be implied from a fully correct answer, but if the answer is not correct, we must see substitution of limits as stated.		

Type MS · Deriving the double-angle formulae

Answer 20 of 24 · Question 10 part a.ii · 3 marks

2023 NOV · Paper 01 · extract 5 of 5 · find part a.ii in this mark scheme extract

Question	Scheme	Marks
	As with many log questions, there are a number of ways of reaching a correct answer. **The first scheme shows 3 common routes, with a set of general principles in the notes.** **Not all variations in methods will have been captured, please mark to general principles, referring ANY work where this is not possible and worthy of marks to REVIEW.** **This includes work that converts everything to completely new bases.**	

Type MS · Deriving the double-angle formulae

Answer 21 of 24 · Question 10 part a.i · 0 marks

2025 JUN · Paper 01 (regional R) · extract 1 of 4 · find part a.i in this mark scheme extract

10 (a)(i)	$\cos(A+A)=\cos A\cos A-\sin A\sin A\rightarrow\cos 2A=\cos^2 A-\sin^2 A$ $\cos 2A=1-\sin^2 A-\sin^2 A\Rightarrow\cos 2A=1-2\sin^2 A$ * $\cos 2A-1=-2\sin^2 A$ or $1-\cos 2A=2\sin^2 A$ $\sin^2 A=\frac{1-\cos 2A}{2}$ *	M1 M1 A1*cso [3]
/ALT (a)(i)	Starting from r.h.s of equation: $\cos(A+A)=\cos A\cos A-\sin A\sin A\rightarrow\cos 2A=\cos^2 A-\sin^2 A$ Gives $\frac{1-\cos 2A}{2}=\frac{1-(\cos^2 A-\sin^2 A)}{2}$ $=\frac{\sin^2 A+\sin^2 A}{2}=\sin^2$ as required.	M1 M1 A1*cso [3]
(ii)	$\left(\cos 2A=\cos^2 A-\sin^2 A\right.$ or $\left.\cos 2A=1-2\sin^2 A\right)$ $\cos 2A=1-\sin^2 A-\sin^2 A$ or $\cos 2A=1-2(1-\cos^2 A)$ $\rightarrow\cos 2A=2\cos^2 A-1$ $\cos 2A+1=2\cos^2 A$ $\cos^2 A=\frac{\cos 2A+1}{2}$ *	M1 A1*cso [2]
ALT (ii)	Starting from r.h.s of equation: $\cos 2A=\cos^2 A-\sin^2 A$ Gives $\frac{\cos 2A+1}{2}=\frac{\cos^2 A-\sin^2 A+1}{2}$ $=\frac{\cos^2 A+\cos^2 A}{2}=\cos^2 A$ as required.	 M1 A1*cso [2]
(b)	$\cos 2x=\sin x$ $1-2\sin^2 x=\sin x\rightarrow 0=2\sin^2 x+\sin x-1$ $0=(2\sin x-1)(\sin x+1)$ $\sin x=\frac{1}{2},(-1)\rightarrow x=$ $\left(\frac{\pi}{6},\frac{1}{2}\right)$	M1 M1 M1 A1 A1 [5]

Type MS · Deriving the double-angle formulae

Answer 21 of 24 · Question 10 part a.i · 0 marks

2025 JUN · Paper 01 (regional R) · extract 2 of 4 · find part a.i in this mark scheme extract

(c)	$\pi\int_0^{\frac{\pi}{6}}\left(\cos^2 2x-\sin^2 x\right)dx$	B1ft
	$(\pi)\int_0^{\frac{\pi}{6}}\left(\frac{\cos 4x+1}{2}\right)-\left(\frac{1-\cos 2x}{2}\right)dx$	M1
	$\left[(\pi)\int_0^{\frac{\pi}{6}}\left(\frac{\cos 4x}{2}+\frac{\cos 2x}{2}\right)dx\right]$	
	$(\pi)\left[\frac{\sin 4x}{8}+\frac{\sin 2x}{4}\right]_0^{\frac{\pi}{6}}$	M1
	$(\pi)\left(\frac{\sin 4\left(\frac{\pi}{6}\right)}{8}+\frac{\sin 2\left(\frac{\pi}{6}\right)}{4}\right)$	M1
	$\frac{3\sqrt{3}}{16}\pi$	A1 [5]
		Total 15 marks

Type MS · Deriving the double-angle formulae

Answer 21 of 24 · Question 10 part a.i · 0 marks

2025 JUN · Paper 01 (regional R) · extract 3 of 4 · find part a.i in this mark scheme extract

Part	Mark	Additional Guidance
10 (a)(i)	M1	Correct use of the addition formula as shown leading to $\cos 2A = \cos^2 A - \sin^2 A$
	M1	Correct substitution of $1-\sin^2 A$ to get $\cos 2A = 1 - 2\sin^2 A$
	A1*cso	Fully correct solution with the minimum steps shown, no errors.
ALT (a)(i)	M1	Correct use of the addition formula as shown leading to $\cos 2A = \cos^2 A - \sin^2 A$
	M1	Correct substitution into the right hand side of the equation.
	A1*cso	Rearrangement of resultant expression to reach left hand side of original equation., without errors..
(ii)	M1	Replacement of $\sin^2 A$ with $1-\cos^2 A$ in an expression from their part (i)
	A1*cso	Rearrangement of a correct equation to attain the given result, minimum steps shown, no errors.
ALT (ii)	M1	Replacement pf cos2A with expression from (i)
	A1*cso	Rearrangement of resultant expression to reach left hand side of original equation, without errors.
(b)	M1	Correctly equates the 2 expressions
	M1	Correctly uses the identity for cos2x and rearranges to an equation of the form $0 = a\sin^2 x + b\sin x + c \qquad a,b,c \neq 0$
	M1	Uses a full method to solve their equation of the form $0 = a\sin^2 x + b\sin x + c \qquad a,b,c \neq 0$ to find the inverse sine to arrive at a value for x. See general guidance for the solution of a 3TQ.
	A1	For the correct value of x
	A1	For correct coordinates, allow omission of brackets or x = and y = . **SC**: Allow A1 where candidate works in degrees and gets x = 30 degrees
(c)	B1ft	For the integral shown, ft their limits only.
	M1	An attempt to replace $\cos^2 2x$ **and** $\sin^2 x$ with an identity from part (a). At least one replacement must be fully correct.
	M1	At least one term correctly integrated.
	M1	Substitution of their limits clearly seen into a changed expression. This can be implied by a correct answer. Where the expression is not correct, or their limits are not correct, the substitution must be seen into at least half of their terms.
	A1	For the correct answer. Withold this mark if the candidate works in degrees rather than radians

Type MS · Deriving the double-angle formulae

Answer 21 of 24 · Question 10 part a.i · 0 marks

2025 JUN · Paper 01 (regional R) · extract 4 of 4 · find part a.i in this mark scheme extract

Question number	Scheme	Marks

Type MS · Deriving the double-angle formulae

Answer 22 of 24 · Question 10 part a.ii · 5 marks

2025 JUN · Paper 01 (regional R) · extract 1 of 4 · find part a.ii in this mark scheme extract

10 (a)(i)	$\cos(A+A)=\cos A\cos A-\sin A\sin A\rightarrow\cos 2A=\cos^2 A-\sin^2 A$ $\cos 2A=1-\sin^2 A-\sin^2 A\Rightarrow\cos 2A=1-2\sin^2 A$ * $\cos 2A-1=-2\sin^2 A$ or $1-\cos 2A=2\sin^2 A$ $\sin^2 A=\frac{1-\cos 2A}{2}$ *	M1 M1 A1*cso [3]
/ALT (a)(i)	Starting from r.h.s of equation: $\cos(A+A)=\cos A\cos A-\sin A\sin A\rightarrow\cos 2A=\cos^2 A-\sin^2 A$ Gives $\frac{1-\cos 2A}{2}=\frac{1-(\cos^2 A-\sin^2 A)}{2}$ $=\frac{\sin^2 A+\sin^2 A}{2}=\sin^2$ as required.	M1 M1 A1*cso [3]
(ii)	$\left(\cos 2A=\cos^2 A-\sin^2 A\right.$ or $\left.\cos 2A=1-2\sin^2 A\right)$ $\cos 2A=1-\sin^2 A-\sin^2 A$ or $\cos 2A=1-2(1-\cos^2 A)$ $\rightarrow\cos 2A=2\cos^2 A-1$ $\cos 2A+1=2\cos^2 A$ $\cos^2 A=\frac{\cos 2A+1}{2}$ *	M1 A1*cso [2]
ALT (ii)	Starting from r.h.s of equation: $\cos 2A=\cos^2 A-\sin^2 A$ Gives $\frac{\cos 2A+1}{2}=\frac{\cos^2 A-\sin^2 A+1}{2}$ $=\frac{\cos^2 A+\cos^2 A}{2}=\cos^2 A$ as required.	 M1 A1*cso [2]
(b)	$\cos 2x=\sin x$ $1-2\sin^2 x=\sin x\rightarrow 0=2\sin^2 x+\sin x-1$ $0=(2\sin x-1)(\sin x+1)$ $\sin x=\frac{1}{2},(-1)\rightarrow x=$ $\left(\frac{\pi}{6},\frac{1}{2}\right)$	M1 M1 M1 A1 A1 [5]

Type MS · Deriving the double-angle formulae

Answer 22 of 24 · Question 10 part a.ii · 5 marks

2025 JUN · Paper 01 (regional R) · extract 2 of 4 · find part a.ii in this mark scheme extract

(c)	$\pi\int_0^{\frac{\pi}{6}}\left(\cos^2 2x-\sin^2 x\right)dx$	B1ft
	$(\pi)\int_0^{\frac{\pi}{6}}\left(\frac{\cos 4x+1}{2}\right)-\left(\frac{1-\cos 2x}{2}\right)dx$	M1
	$\left[(\pi)\int_0^{\frac{\pi}{6}}\left(\frac{\cos 4x}{2}+\frac{\cos 2x}{2}\right)dx\right]$	
	$(\pi)\left[\frac{\sin 4x}{8}+\frac{\sin 2x}{4}\right]_0^{\frac{\pi}{6}}$	M1
	$(\pi)\left(\frac{\sin 4\left(\frac{\pi}{6}\right)}{8}+\frac{\sin 2\left(\frac{\pi}{6}\right)}{4}\right)$	M1
	$\frac{3\sqrt{3}}{16}\pi$	A1 [5]
		Total 15 marks

Type MS · Deriving the double-angle formulae

Answer 22 of 24 · Question 10 part a.ii · 5 marks

2025 JUN · Paper 01 (regional R) · extract 3 of 4 · find part a.ii in this mark scheme extract

Part	Mark	Additional Guidance
10 (a)(i)	M1	Correct use of the addition formula as shown leading to $\cos 2A = \cos^2 A - \sin^2 A$
	M1	Correct substitution of $1-\sin^2 A$ to get $\cos 2A = 1 - 2\sin^2 A$
	A1*cso	Fully correct solution with the minimum steps shown, no errors.
ALT (a)(i)	M1	Correct use of the addition formula as shown leading to $\cos 2A = \cos^2 A - \sin^2 A$
	M1	Correct substitution into the right hand side of the equation.
	A1*cso	Rearrangement of resultant expression to reach left hand side of original equation., without errors..
(ii)	M1	Replacement of $\sin^2 A$ with $1-\cos^2 A$ in an expression from their part (i)
	A1*cso	Rearrangement of a correct equation to attain the given result, minimum steps shown, no errors.
ALT (ii)	M1	Replacement pf cos2A with expression from (i)
	A1*cso	Rearrangement of resultant expression to reach left hand side of original equation, without errors.
(b)	M1	Correctly equates the 2 expressions
	M1	Correctly uses the identity for cos2x and rearranges to an equation of the form $0 = a\sin^2 x + b\sin x + c \qquad a,b,c \neq 0$
	M1	Uses a full method to solve their equation of the form $0 = a\sin^2 x + b\sin x + c \qquad a,b,c \neq 0$ to find the inverse sine to arrive at a value for x. See general guidance for the solution of a 3TQ.
	A1	For the correct value of x
	A1	For correct coordinates, allow omission of brackets or x = and y = . **SC**: Allow A1 where candidate works in degrees and gets x = 30 degrees
(c)	B1ft	For the integral shown, ft their limits only.
	M1	An attempt to replace $\cos^2 2x$ **and** $\sin^2 x$ with an identity from part (a). At least one replacement must be fully correct.
	M1	At least one term correctly integrated.
	M1	Substitution of their limits clearly seen into a changed expression. This can be implied by a correct answer. Where the expression is not correct, or their limits are not correct, the substitution must be seen into at least half of their terms.
	A1	For the correct answer. Withold this mark if the candidate works in degrees rather than radians

Type MS · Deriving the double-angle formulae

Answer 22 of 24 · Question 10 part a.ii · 5 marks

2025 JUN · Paper 01 (regional R) · extract 4 of 4 · find part a.ii in this mark scheme extract

Question number	Scheme	Marks

Type MS · Deriving the double-angle formulae

Answer 23 of 24 · Question 11 part a · 3 marks

2025 NOV · Paper 01 · find part a in this mark scheme extract

11(a)	$\frac{1-\cos 2A}{1+\cos 2A}=\frac{1-\left(\cos^2 A-\sin^2 A\right)}{1+\left(\cos^2 A-\sin^2 A\right)}$	M1
	$=\frac{1-\left(1-\sin^2 A-\sin^2 A\right)}{1+\left(\cos^2 A-\left(1-\cos^2 A\right)\right)}$	M1
	$=\frac{2\sin^2 A}{2\cos^2 A}=\tan^2 A$ *	A1cso [3]
(b)	$\frac{3-3\cos 4x}{1+\cos 4x}+\frac{5\sin 2x}{\cos 2x}=2\Rightarrow 3\tan^2 2x+5\tan 2x-2=0$	M1,M1
	$\Rightarrow\left(3\tan 2x-1\right)\left(\tan 2x+2\right)=0\Rightarrow\tan 2x=\frac{1}{3},-2$	M1A1
	$\Rightarrow 2x=-161.565, 18.43, -63.43, 116.56$	M1
	$\Rightarrow x=-80.8,\ -31.7,\ 9.2,\ 58.3$	A1A1
	Accept awrt 1dp	[7]
		Total 10 marks

Type MS · Deriving the double-angle formulae

Answer 24 of 24 · Question 8 part a · 2 marks

2024 NOV · Paper 01 · find part a in this mark scheme extract

Number		
8 (i) (a)	$\left(\tan 2A = \tan(A+A) =\right)\dfrac{\tan A+\tan A}{1-\tan A\tan A}$ $=\dfrac{2\tan A}{1-\tan^2 A}$ *	M1 A1cso* [2]
(b)	$\tan A-\dfrac{2\tan A}{1-\tan^2 A}=0$ $\tan A(1-\tan^2 A)-2\tan A=0$ $\tan A-\tan^3 A-2\tan A=0$ $\tan^3 A+\tan A=0$ $\tan A(\tan^2 A+1)=0$ $\tan A=0 \qquad A=0, 180$ $\left(\tan^2 A=-1 \qquad \text{no solutions}\right)$	M1 M1 M1 A1, A1 [5]
ALT	$\tan A-\dfrac{2\tan A}{1-\tan^2 A}=0$ $\tan A\left(1-\dfrac{2}{1-\tan^2 A}\right)=0 \Rightarrow 1-\tan^2 A-2=0 \text{ or } 2=1-\tan^2 A$ $\tan^2 A=-1$ $\tan A=0 \qquad A=0, 180$ $\left(\tan^2 A=-1 \qquad \text{no solutions}\right)$	
(ii)	$\cos x\cos\dfrac{\pi}{6}+\sin x\sin\dfrac{\pi}{6}\left(=\sin x\right)$ $\sin x+\sqrt{3}\cos x=2\sin x \Rightarrow \sqrt{3}=\tan x$ $x=-\dfrac{2\pi}{3},\dfrac{\pi}{3},\dfrac{4\pi}{3}$	M1 dM1 A1 A1 [4]
		Total 11 marks